

NENA Standard for the Conveyance of Emergency Incident Data Objects (EIDOs) between Next Generation (NG9-1-1) Systems and Applications

Abstract: Standard defining how EIDOs are conveyed between Functional Elements



NENA Standard for the Conveyance of Emergency Incident Data Objects (EIDOs) between Next Generation (NG9-1-1) Systems and Applications

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1 Executive Overview

The Emergency Incident Data Object (EIDO) is the JavaScript Object Notation (JSON) object that contains incident and related information intended to be passed between Functional Elements (FEs) for NENA-STA-021's [14] stated purpose of processing emergency incidents, such as for use with systems handling 9-1-1 calls. EIDO itself constitutes a comprehensive mechanism for passing needed critical information fields between FEs and subsequently incident processing stakeholders. Without an equally reliable method of transport, EIDOs would be left stranded inside proprietary boxes or on disparate islands.

It is the purpose of the Standard for the Conveyance of Emergency Incident Data Objects to define a transport model for the secure, verified, and reliable delivery of EIDOs, both during the processing of emergency calls and thereafter. This version of EIDO Conveyance provides for mechanisms for push notifications and requests for resources.

Users involved in the exchange of incident information that use the EIDO to accomplish that exchange may be interested in this document's described mechanisms for conveyance of EIDOs. EIDO Conveyance provides a crucial transport mechanism that offers the needed solution for passing an EIDO to authorized subscribers based on their incident scope or needs. For example, two FEs such as Incident Record Handling FEs and an Incident Data eXchange (IDX) would use subscriptions to share EIDOs on behalf of the users working on incidents related to calls handled by them. By defining a secure and reliable transport model for FEs to send EIDOs, EIDO Conveyance solves the problem of how to send the needed information to FEs that may not utilize the same vendor or function.

The reader should be familiar with general concepts of HTTP, JSON, and WebSocket to effectively understand this document and referenced concepts.

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2 Document Conventions

NENA: The 9-1-1 Association improves 9-1-1 through research, standards development, training, education, outreach, and advocacy. Our vision is a public made safer and more secure through universally-available state-of-the-art 9-1-1 systems and better-trained 9-1-1 professionals. Learn more at nena.org.

2.1 Document Terminology

This section defines keywords, as they should be interpreted in NENA documents. The form of emphasis (UPPER CASE) shall be consistent and exclusive throughout the document. Any of these words used in lower case and not emphasized do not have special significance beyond normal usage.

1. **MUST, SHALL, REQUIRED:** These terms mean that the definition is a normative (absolute) requirement of the specification.
2. **MUST NOT:** This phrase, or the phrase "SHALL NOT", means that the definition is an absolute prohibition of the specification.
3. **SHOULD:** This word, or the adjective "RECOMMENDED", means that there may exist valid reasons in particular circumstances to ignore a particular item, but the full implications must be understood and carefully weighed before choosing a different course.
4. **SHOULD NOT:** This phrase, or the phrase "NOT RECOMMENDED" means that there may exist valid reasons in particular circumstances when the particular behavior is acceptable or even useful, but the full implications should be understood and the case carefully weighed before implementing any behavior described with this label.
5. **MAY:** This word, or the adjective "OPTIONAL", means that an item is truly optional. One vendor may choose to include the item because a particular marketplace requires it or because the vendor feels that it enhances the product while another vendor may omit the same item. An implementation which does not include a particular option "must" be prepared to interoperate with another implementation which does include the option, though perhaps with reduced functionality. In the same vein an implementation which does include a particular option "must" be prepared to interoperate with another implementation which does not include the option (except, of course, for the feature the option provides.)

These definitions are based on RFC 2119 [5].

2.2 Describing Interfaces for XML and JSON Objects

This document defines one or more interfaces, often called "Application Programming Interfaces" (APIs) that provide a standardized way to communicate with a Functional

Element. The interface consists of data objects and the operations that apply to those data objects. In this document, the operations and data objects are documented as a formal definition of an interface or schema and a set of "business rules" which are also required in the software or database environment to comply with the standard and achieve interoperability. The business rules are specified with normative statements using keywords that are defined in the Document Terminology section above.

2.3 Describing REST/JSON Interfaces

This document uses "OpenAPI" Version 3.1 [26] as the formal description of a REST/JSON interface. This description is manifested in a file ("yaml") that is not contained directly in this document. The file is maintained in a NENA GitHub repository. In the interface description, a hyperlink to the yaml file is provided.

In the description of the interface in the text, a table often appears that describes the parameters to the interface and the data objects the interface exchanges. These tables are informative; the yaml file is the normative description of the interface and was reviewed with the text of this document. If there is a discrepancy between the table and the yaml file, the yaml controls. In the tables, a column is headed "Required". Three values may occur in this column. "y" means the parameter or member is required to be present. It will appear in the "required" section of the yaml file. "n" means that the parameter or member is optional. It will not appear in the "required" section of the yaml file. "c" means that the parameter or member is conditional. This value alerts the reader that a business rule is specified that controls the presence of the parameter or member. It will not appear in the "required" section of the yaml file.

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2.5 Reason for Issue/Reissue

NENA reserves the right to modify this document. Upon revision, the reason(s) will be provided in the table below.

Document Number	Approval Date	Reason for Issue/Reissue
NENA-STA-024.1-2023	January 15, 2023	Initial Document

NENA-STA-024.1a-2023	March 21, 2023	Publishing error corrections: 1) Syntax: wss://<target host>/ent<eido subscribe-path> 2) Example: wss://idx.esinet.net/IncidentData/ents ubscribe 3) The response to the request, logged via a EidoDereferenceFactoryQueryRespon seLogEvent, MUST include the "direction" member, the "queryId" member, and either a "generatedURI" member or, a "responseError" member that contains an error code from the Error Codes Registry <u>and a</u> <u>"responseText"</u> in the case of an error. 4) an "incidentId" member in the header field of the log record, <u>and a</u> <u>"reasonCode"</u> member containing an associated reason code from the EIDO Dereference Denial registry <u>and a</u> <u>"reasonText"</u> member. 5) an "incidentId" member in the header field of the log record, and a "errorCodestatus" member containing a status code from the EIDO Transmission Status registry indicating the resultant failure <u>and an</u> <u>"errorText"</u> member. 6) If the subscribe request was denied the log event MUST include an error message in the "errorCoderesponseError" <u>and</u> <u>"errorText"</u> members <u>respectively</u>.
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Document Number	Approval Date	Reason for Issue/Reissue
		7) a "statusCode" member and a "statusTextDescription" member. 8) and an optional "closeTextReason" member containing a reason per RFC 6455 [10]. 9) See NENA Master Glossary of 9-1-1 Terminology, NENA-ADM-000 Error! Reference source not found. Error! Reference source not found, for a complete listing of terms used in NENA documents. https://idx.esinet.net/IncidentData/v1/incidentspsap.alleghany.pa.us/fee/eide/urn:emergency:uid:incident:6955@psap.alleghany.pa.us 10) GET /IncidentData/v1/fee/dereference/urn:emergency:uid:incident:6955@psap.alleghany.pa.us HTTP/1.1 11) Host: psap.alleghany.pa.us

Document Number	Approval Date	Reason for Issue/Reissue
NENA-STA-024.1b-2023	06/18/2024	Publishing errors found in NG9-1-1 Logging Service yaml: “The yaml for the following LogEvents includes an “incidentId” element in the LogEvent body that conflicts with the “incidentId” that is common to all LogEvents. The text in NENA-STA-024.1a-2023 is fine. The error was introduced when the yaml was created. EidoDereferenceFactoryQueryLogEvent, SubscriptionRequestedLogEvent, SubscriptionTerminatedLogEvent, EidoDeniedLogEvent, EidoTransmissionErrorLogEvent” Revised version at https://github.com/NENA911/Logging-Service/tree/STA-024.1b.
NENA-STA-024.1.1-2025	January 30, 2025	Version 1.1 revision includes methods for resource discovery and push notifications for requests for service, requests for awareness, and responses thereto. Other changes include corrections, clarifications, and other improvements.

3 Emergency Incident Data Object Conveyance

PSAPs will make extensive use of the NENA Standard for Emergency Incident Data Object (EIDO) (NENA-STA-021) [14]. Most interactions between FEs inside a PSAP, between PSAPs, and between PSAPs and other entities, involve sending and receiving an EIDO.

All transmissions of EIDOs MUST use TLS security as per NENA-STA-010[14][13].

The content of an EIDO MUST conform to the data rights management policy of the owner of the data per the Authorization and Data Rights Management section of NENA-STA-010 [13][13].

For interoperability, all implementations SHALL accept EIDOs of any size up to and including 10,485,760 bytes (ten mebibytes).

3.1 Endpoint Discovery

An agency MAY use the Service/Agency Locator (S/AL) to identify the URI for the WebSocket for another agency per NENA-STA-010[14][13][13] based on the agency's name.

An agency MAY use the ECRF with the service URN
urn:emergency:service:serviceagencyLocator.psap per NENA-STA-010[14][13] to discover the primary PSAP at a location (as represented by civic address, a point, or a polygon), or agencies providing a specific service type, e.g.,
"urn:emergency:service:serviceagencyLocator.fire", for a particular location. The query will return the URI of the Service/Agency Locator record for that agency. The record contains a value for eidoInterfaceUri which is the URI of WebSocket to be used. If a polygon is specified then multiple Service/Agency Locator records may be returned.

The URL returned by any of those methods would be the URI for that agency to which a WebSocket can be established, if one does not already exist, and a payload can be sent via the WebSocket.

3.2 WebSocket-based Transport

Clients establish a connection to a service via a WebSocket per RFC 6455 [10] specifying a WebSocket Protocol named "emergency-ent".

Mutual authentication with credentials traceable to the PSAP Credentialing Agency (PCA) is required for all connections.

Clients MAY then use that WebSocket to submit subscription and Push requests.

3.2.1 Client WebSocket Handshake

3.2.1.1 Client WebSocket Negotiation

3.2.1.1.1 Negotiate WebSocket

Open a WebSocket to a service endpoint.

3.2.1.1.2 HTTP Headers

The following HTTP headers are REQUIRED to establish a WebSocket for the WebSocket Protocol “emergency-ent” per RFC 6455, *The WebSocket Protocol* [10]:

Header Name	Value	Description
Connection	“Upgrade”	Directive to change protocol.
Upgrade	“websocket”	Directive to upgrade to the WebSocket protocol.
Host	Hostname	Identification of the host requested.
Sec-WebSocket-Protocol	“emergency-ent1.1, emergency-ent1.0”	Used to identify the version(s) of the message protocol supported by the requestor.
Sec-WebSocket-Key	base64	The value of this header field MUST be a nonce consisting of a randomly selected 16-byte value that has been base64-encoded (see Section 4 of [7]). The nonce MUST be selected randomly for each connection.
Sec-WebSocket-Version	“13”	WebSocket protocol version number.

The following HTTP headers are REQUIRED to be returned in response to the request to establish a WebSocket:

Header Name	Value	Description
Connection:	"Upgrade"	The server is upgrading the connection.
Upgrade:	"websocket"	The server is upgrading the connection to a WebSocket.
Set-WebSocket-Accept	Base64	Sent from the server to the client to confirm that the server is willing to initiate the WebSocket connection.
Sec-WebSocket-Protocol:	"emergency-ent1.1"	Used to identify the message protocol and its version. If the requestor specified only "emergency-ent1.0" then the responder will return "emergency-ent1.0".

Example: WebSocket Establishment Request

```
GET /eido_urn HTTP/1.1
Host: idx.esinet.net
Connection: Upgrade
Upgrade: websocket
Sec-WebSocket-Protocol: emergency-ent1.1, emergency-ent1.0
Sec-WebSocket-Key: dGhlIHhnbXBsZSBub25jZQ==
Sec-WebSocket-Version: 13
```

Example: WebSocket Establishment Response

```
HTTP/1.1 101 Switching Protocols
Upgrade: websocket
Connection: Upgrade
Sec-WebSocket-Accept: s3pPLMBiTxaQ9kYGzzhZRbK+xOo=
Sec-WebSocket-Protocol: emergency-ent1.1
```

3.3 WebSocket-based Incident Data Subscription

A Functional Element (FE) exposes an interface for subscribing to events or messages as matched by the provided filter. Using the `qualFilter`, the service will support a subscription to matching specific target objects that will be sent via a WebSocket.

3.3.1 General Description

Once the WebSocket has been established, the client may request one or more subscriptions to incident or resource data based on one of the available methods: a subscription that returns one EIDO for all new incidents or a filtered subset thereof (New Incidents), a subscription for all EIDOs for a specific incident or rate-limited subset thereof

(Single Incident), or a subscription for all EIDOs for a specific resource or rate-limited subset thereof (Resource).

For interoperability, all implementations SHALL accept subscription requests of any size up to and including 65,536 bytes.

All communication related to a subscription is handled over the WebSocket.

Upon successful subscription, notifications SHALL be sent to the subscriber as they match the criteria associated with the subscription until the subscription expires or an unsubscribe occurs.

For a New Incidents subscription, a notification SHALL be sent immediately that contains all active incidents that match the criteria. If no incidents match the criteria, an empty notification SHALL be sent as described in the minRate section below.

For a Single Incident subscription, a notification SHALL be sent immediately that contains that incident. A Single Incident subscription SHALL be accepted for a closed incident if the request is made within five minutes of its closure, in which case a single EIDO representing the last state of that incident SHALL be sent and no further notifications will be made for that incident; however, a notifier MAY accept a Single Incident subscription for an incident that has been closed longer than five minutes. If a closed incident is reopened and that incident matches the criteria for a New Incidents subscription as described in the preceding two paragraphs, a notification SHALL be sent.

For a Resource subscription, a notification SHALL be sent immediately that contains that resource. A notification SHALL be sent for every change of state of the resource subject to rate constraints.

A notifier will not send any notifications until a successful subscription has taken place. Notifications will only be sent if they match the filter or other subscription criteria.

A subscriber SHALL send subsequent subscribe actions to reset the expiry. To reset the expiry a request MUST contain the subscription identifier provided when originally subscribing. The new expiration is provided back in the response. An empty notification is used as a keep-alive mechanism for long term subscriptions.

Any change to a subscription by a subscriber other than resetting the expiry requires creating a new subscription and then sending an unsubscribe message for the old subscription.

Once established, a WebSocket SHALL handle multiple subscriptions.

Upon expiration of the last subscription still in effect the notifier will close the WebSocket. It is up to the subscriber to detect this closure and act appropriately.

If the WebSocket is closed (including if the underlying TCP socket is closed) then the notifier shall flush all subscriptions established using that WebSocket without notifying the subscriber and the subscriber shall reopen the WebSocket and reestablish any subscriptions still desired as new subscriptions.

3.3.2 URI scheme

A URI SHALL exist for each Functional Element that is an EIDO notifier.

The external subscription URI for an agency or service is discoverable using the service/agency locator as specified in STA-010 [13].

The Secure WebSocket URI Scheme is used for subscriptions per RFC 3986 [6].

Multiple notifier functional elements can share the same subscription interface, the implementation of which is outside the scope of this standard.

Syntax: `wss://<target host>/<eido subscribe path>`

Example: `wss://idx.esinet.net/IncidentData/subscribe`

3.3.3 Client WebSocket API Actions

Following the establishment of the WebSocket, the client sends messages to the FE using WebSocket text frames, which are encoded using UTF-8. The methods defined herein are subscribe and unsubscribe.

Payload Object	Description
<code>subscribe:{<subscription_object>}</code>	Apply an EIDO subscription for specific results. A subscription ID is provided back in an acknowledgment and used to keep a subscription alive or make subsequent filter changes.
<code>unsubscribe:{<subscriptionId>}</code>	Unsubscribe from the given subscription ID.

If a request is made for any method other than subscribe or unsubscribe then the notifier shall respond with the following:

```
{
  "response": {
    "requestId": "63bfd612-f558-4489-b653-caa76a7edb9b",
    "statusCode": 501,
    "statusText": "Not implemented"
  }
}
```

If no requestId element is specified in the request, no requestId element will be included in the response.

3.3.4 Objects

There are eight types of conveyance objects. They include:

1. event
2. eventResponse
3. subscribe
4. subscribeResponse
5. terminate
6. terminateResponse
7. unsubscribe
8. unsubscribeResponse

The followings sections provide more detail to describe the properties present in each of these objects.

3.3.4.1 subscribe

3.3.4.1.1 client request

Name	Type	Required	Description
requestId	string	Y	An ID generated by the subscriber that will be returned by the notifier in its response to the request.
requestType	string	C	Type of request, e.g., "EIDO".
requestSubType	string	C	The requestSubType provides for different actions based on the requestType specified.

Name	Type	Required	Description
requestAccepts	string	C	The MIME type associated with the requestType, which, for EIDOs, is application/emergency.eido+json, and the major versions thereof accepted in the notification payload as a quoted comma-delimited list, as in the Accept HTTP header per RFC 7231 [11], e.g., application/emergency.eido+json; version="1" or application/emergency.eido+json; version="1,2,3". The major version is required and is the only extension parameter permitted. Any minor version associated with the major version MAY be sent.
qualFilter	string	C	When subscribing for new incidents, the qualFilter may be used to be notified of a subset thereof.
qualFilterEvalPeriod	integer	C	If qualFilter specified, a positive integer specifying the requested number of seconds between qualFilter evaluations.
minRate	integer	C	Number of seconds after the last notification after which a notification must be sent consistent with RFC 6446 [9][9]. If not specified there is no minimum rate.
maxRate	integer	C	Minimum time in seconds to wait between notifications which is the maximum rate at which notifications are sent consistent with RFC 6446 [9]. If not specified there is no maximum rate.
subscriptionId	string	N	The subscriptionId previously returned by the notifier in the subscribeResponse. This is used for keeping a subscription alive.
incidentId	string	C	The Incident Tracking Identifier associated with the call.
unitId	string	C	The unit ID associated with the resource.

Name	Type	Required	Description
expires	integer	N	Number of seconds after which the subscription shall expire.

The requestType element is REQUIRED for a new subscription (i.e., if no subscriptionId element provided), otherwise, MUST be omitted.

If the requestType element is omitted, the requestSubType element MUST be omitted. If the value of the requestType element is "EIDO", the requestSubType element MUST be present and its value MUST be one of "single", "new", or "resource". Other request types may have their own conditions or may not require a requestSubType element.

The requestAccepts element is REQUIRED for a new subscription, i.e., if the subscriptionId element is not specified.

The qualFilter element MUST NOT be specified if an incidentId or a unitId is specified. The qualFilter element is OPTIONAL if the value of the requestType element is "EIDO" and neither the incidentId nor the unitId element is specified.

The qualFilterEvalPeriod element is OPTIONAL if the qualFilter element is specified, otherwise the qualFilterEvalPeriod element MUST NOT be specified.

The minRate element is OPTIONAL if the subscriptionId element is not specified, otherwise the minRate element MUST NOT be specified.

The maxRate element is OPTIONAL if the subscriptionId element is not specified, otherwise the maxRate element MUST NOT be specified.

The incidentId element MUST be present to subscribe to a specific incident when the value of the requestType element is "EIDO" and the value of the requestSubType element is "single" in order to subscribe to a specific incident. The incidentId element MUST not be present for other values of a requestSubType element.

The unitId element MUST be present to subscribe to a specific resource when the value of the requestType element is "EIDO" and the value of the requestSubType element is "resource" in order to subscribe to a specific resource. The unitId element MUST not be present for other values of a requestSubType element.

client subscription request filter example:

```
{
  "subscribe": {
    "requestId": "63bfd612-f558-4489-b653-caa76a7edb9b",
    "requestType": "EIDO",
    "requestSubType": "new",
    "requestAccepts": "application/emergency.eido+json;
version=\\"1,2,3\\"";
    "qualFilter": "<filter_string>",
    "minRate": 1500
  }
}
```

client reset expire timer example:

```
{
  "subscribe": {
    "requestId": "63bfd612-f558-4489-b653-caa76a7edb9b",
    "subscriptionId": "sub1234"
  }
}
```

client subscription request for an incidentId example:

```
{
  "subscribe": {
    "requestId": "63bfd612-f558-4489-b653-caa76a7edb9b",
    "requestType": "EIDO",
    "requestSubType": "single",
    "requestAccepts": "application/emergency.eido+json;
version=\\"1,2,3\\"";
    "incidentId":
    "urn:emergency:uid:incidentid:a56e556d871:bcf.state.pa.us"
  }
}
```

Note – The request may require an identifier related to a discrete object. For EIDO, incidentId is used to subscribe to a specific incident ID. For any other transport use case it is an event level identifier as defined by that specific protocol standard, e.g., Service or Element State Event notification it can be the SIP Call-ID of the SUBSCRIBE/NOTIFY dialog.

3.3.4.1.2Subscribe Response - “subscribe” service response

3.3.4.1.2.1 subscribeResponse

Name	Type	Required	Description
requestId	string	Y	The ID provided by the subscriber.
subscriptionId	string	C	The globally unique identifier for the subscription.
qualFilterEvalPeriod	integer	C	Negotiated number of seconds between qualFilter evaluations. If requested, may be a longer period than the request but not shorter.
minRate	integer	C	May be a longer period than requested but not shorter.
maxRate	integer	C	May be a shorter period than requested but not longer.
expires	integer	C	Subscription period in seconds relative to the notifier’s establishment or renewal of the subscription.
statusCode	integer	Y	The statusCode element is a 3-digit integer result code of the attempt to understand and satisfy the request as defined in the Hypertext Transfer Protocol Status Code Registry (RFC 7231) [11] or the IANA Status Codes Registry [12][12].
statusText	string	Y	The statusText provides a short textual description related to the statusCode.

The subscriptionId element is REQUIRED if subscription is successful.

The qualFilterEvalPeriod element is REQUIRED if the qualFilter element is specified.

The minRate element is REQUIRED if the minRate element was specified by the subscriber.

The maxRate element is REQUIRED if the maxRate element was specified by the subscriber.

The expires element is REQUIRED if subscription is successful.

The response to the subscription request SHALL contain one of the following status codes:

Code	Description	Subscriber Action
200	OK	None
400	Bad Request	None
403	Forbidden	None
404	Not Found	None
406	Unacceptable	None
408	Request Timeout	None
413	Payload too large	None
481	Subscription does not exist	None
500	Internal Server Error	None
503	Service Unavailable	Retry request

Successful "subscribe" service response example:

```
{
  "subscribeResponse": {
    "requestId": "63bfd612-f558-4489-b653-caa76a7edb9b",
    "subscriptionId": "sub1234",
    "expires": 3600,
    "statusCode": 200,
    "statusText": "OK"
  }
}
```

Unsuccessful "subscribe" service response example:

```
{
  "subscribeResponse": {
    "requestId": "63bfd612-f558-4489-b653-caa76a7edb9b",
    "statusCode": 481,
    "statusText": "Subscription does not exist"
  }
}
```

3.3.4.2 unsubscribe

3.3.4.2.1 client request

Name	Type	Required	Description
requestId	string	Y	An ID generated by the subscriber that will be returned by the notifier in its response to the request.
subscriptionId	string	Y	The subscriptionId sent by the service. This is provided back in an acknowledgment by the service, used for keeping a subscription alive or subsequent filter modifications.

Client unsubscribe example:

```
{
  "unsubscribe": {
    "requestId": "63bfd612-f558-4489-b653-caa76a7edb9b",
    "subscriptionId": "sub1234"
  }
}
```

3.3.4.2.2 "unsubscribe" service response

3.3.4.2.2.1 unsubscribeResponse

Name	Type	Required	Description
requestId	string	Y	The ID specified by the subscriber.
subscriptionId	string	Y	Globally unique identifier for the subscription.
statusCode	integer	Y	Status code of response according to the request, e.g., 200, as defined in the Hypertext Transfer Protocol Status Code Registry (RFC 7231) [11] or the IANA Status Codes Registry [12].
statusText	string	Y	The statusText is intended to provide information related to the statusCode.

The response to the unsubscribe request SHALL contain one of the following statusCode values:

Code	Description	Subscriber Action
200	OK	None
400	Bad Request	None
408	Request Timeout	None
413	Payload too large	None
481	Subscription does not exist	None
500	Internal Server Error	None
503	Service Unavailable	Retry request

Successful "unsubscribe" service response:

```
{
  "unsubscribeResponse": {
    "subscriptionId": "sub1234",
    "statusCode": 200,
    "statusText": "Unsubscribed"
  }
}
```

Unsuccessful "unsubscribe" service response:

```
{
  "unsubscribeResponse": {
    "subscriptionId": "sub1234",
    "statusCode": 481,
    "statusText": "Subscription Not Found"
  }
}
```

3.3.5 Notification Model

Based on the subscription, qualifying incidents will be sent to the subscriber in the form of a notification that contains an event. The subscriber will respond with an eventResponse. A notifier sends a terminate message when a subscription is terminated such as due to subscription expiration. An explicit unsubscribe request by the subscriber will not cause the notifier to generate a terminate message.

3.3.5.1 Notification Objects

Payload Object	Description
event:{<event_object>}	Notification has been received.
terminate:{<subscriptionId>}	Notification of termination of a subscription.

3.3.5.2 Event Object

Name	Type	Required	Description
subscriptionId	string	Y	Subscription to which this event belongs.
transactionId	string	Y	A GUID used to identify a notification for acknowledgment.
notification	array	Y	Array of objects, e.g., EIDOs.

An example of a notification is:

```
{
  "event": {
    "subscriptionId": "sub1234",
    "transactionId": "f894fjhdsh940jdf@192.168.2.1",
    "notification": [
      <EIDO>
    ]
  }
}
```

3.3.5.3 Event Notification Response

3.3.5.3.1 Event Response

Name	Type	Required	Notes
transactionId	string	Y	GUID used to identify a transaction for acknowledgment.
statusCode	integer	Y	Status code of response according to the request, e.g., 200, as defined in the Hypertext Transfer Protocol Status Code Registry (RFC 7231) [11] or the IANA Status Codes Registry [12].
statusText	string	Y	The statusText is intended to provide information related to the statusCode.

One of the following statusCode values MUST be returned.

Code	Description	Notifier Action
200	OK	None
400	Bad Request	None
404	Not Found	None
408	Request Timeout	None
413	Payload too large	Send an event for each notification object without regard to rate limiting
481	Subscription does not exist	Do not send notifications for that subscription
500	Internal Server error	None
501	Not implemented by subscriber	None
503	Service Unavailable	Retry

3.3.5.3.2Subscriber Notification Response example:

```
{
  "eventResponse": {
    "transactionId": "f894fjhdsh940jdf@192.168.2.1"
    "statusCode": 200,
    "statusText": "OK"
  }
}
```

3.3.5.4 Terminate Object

Name	Type	Required	Description
requestId	string	Y	An ID generated by the notifier that will be returned by the subscriber in its response to the request.
subscriptionId	string	Y	Subscription being terminated.

An example of a termination is:

```
{
  "terminate": {
    "requestId": "63bfd612-f558-4489-b653-caa76a7edb9b",
    "subscriptionId": "sub1234"
  }
}
```

3.3.5.5 Terminate Notification Response

3.3.5.5.1 Terminate Response

Name	Type	Required	Notes
requestId	string	Y	The ID generated by the notifier.
subscriptionId	string	Y	Subscription being terminated.
statusCode	integer	Y	Status code of response according to the request, e.g., 200, as defined in the Hypertext Transfer Protocol Status Code Registry (RFC 7231) [11] or the IANA Status Codes Registry [12].
statusText	string	Y	The statusText is intended to provide information related to the statusCode.

The response to the terminate request SHALL contain one of the following statusCode values.

Code	Description	Notifier Action
200	OK	None
400	Bad Request	None
408	Request Timeout	None
481	Subscription does not exist	Do not send notifications for that subscription

Code	Description	Notifier Action
500	Internal Server error	None
501	Not implemented by subscriber	None
503	Service Unavailable	Retry

3.3.5.5.2 Terminate Response example:

```
{
  "terminateResponse": {
    "requestId": "63bfd612-f558-4489-b653-caa76a7edb9b",
    "subscriptionId": "sub1234",
    "statusCode": 200,
    "statusText": "OK"
  }
}
```

3.3.6 Use Cases

Use Case	Response
Notifier receives a subscribe message that does not contain a subscriptionId. <pre>{ "subscribe": { "requestId": "63bfd612-f558-4489-b653-caa76a7edb9b", "requestType": "EIDO", "requestSubType": "new", "requestAccepts": "application/emergency.eido+json; version=\\"1,2,3\\"\"", "qualFilter": "<filter_string>", "minRate": 1500 } }</pre>	The subscription is accepted by the notifier. <pre>{ "subscribeResponse": { "requestId": "63bfd612-f558-4489-b653-caa76a7edb9b", "subscriptionId": "sub1234", "expires": 3600, "qualFilterEvalPeriod": 10, "minRate": 1500, "statusCode": 200, "statusMessage": "Subscription Accepted" } }</pre>

Use Case	Response
Notifier receives a subscribe message that contains a subscriptionId but the subscriptionId does not match an active subscriptionId on the established WebSocket. <pre>{ "subscribe": { "requestId": "63bfd612-f558-4489-b653-caa76a7edb9b", "subscriptionId": "sub4321", "minRate": 1500 } }</pre>	Notifier returns a subscribeResponse message with a statusCode of 481 indicating that the subscription was rejected. <pre>{ "subscribeResponse": { "requestId": "63bfd612-f558-4489-b653-caa76a7edb9b", "statusCode": 481, "statusMessage": "Subscription not found" } }</pre>
Notifier receives a subscribe message that contains a subscriptionId that matches the active subscriptionId. The filter contained within the subscribe message is different than the active filter. <pre>{ "subscribe": { "requestId": "63bfd612-f558-4489-b653-caa76a7edb9b", "subscriptionId": "sub1234", "qualFilter": "<filter_string>", "minRate": 1500 } }</pre>	Notifier returns a successful subscribe response message. <pre>{ "subscribeResponse": { "requestId": "63bfd612-f558-4489-b653-caa76a7edb9b", "subscriptionId": "sub1234", "expires": 3600, "qualFilterEvalPeriod": 10, "minRate": 1500, "statusCode": 200, "statusMessage": "Subscription accepted" } }</pre> The updated filter replaces the existing active filter.

Use Case	Response
Notifier receives a subscribe message for an incident ID that is unknown. <pre>{ "subscribe": { "requestId": "63bfd612-f558-4489-b653-caa76a7edb9b", "requestType": "EIDO", "requestSubType": "single", "requestAccepts": "application/emergency.eido+json; version=\\"1,2,3\\"\"", "incidentId": "urn:emergency:uid:incidentid:a56e556d871:bcf.state.pa.us" } }</pre>	Notifier returns a subscribeResponse message with a statusCode of 404 indicating that the subscription was rejected because of the unknown incident. <pre>{ "subscribeResponse": { "requestId": "63bfd612-f558-4489-b653-caa76a7edb9b", "statusCode": 404, "statusMessage": "Incident Does Not Exist" } }</pre>
Notifier receives a subscribe message that contains a unitId that is valid. <pre>{ "subscribe": { "requestId": "63bfd612-f558-4489-b653-caa76a7edb9b", "requestType": "EIDO", "requestSubType": "resource", "requestAccepts": "application/emergency.eido+json; version=\\"1,2,3\\"\"", "unitId": "urn:emergency:uid:unitid:1a12:bcf.state.pa.us" } }</pre>	The subscription is accepted by the notifier. <pre>{ "subscribeResponse": { "requestId": "63bfd612-f558-4489-b653-caa76a7edb9b", "subscriptionId": "sub1234", "expires": 3600, "statusCode": 200, "statusMessage": "Subscription Accepted" } }</pre>

Use Case	Response
Notifier receives a subscribe message that contains a unitId that is invalid. <pre>{ "subscribe": { "requestId": "63bfd612-f558-4489-b653-caa76a7edb9b", "requestType": "EIDO", "requestSubType": "resource", "requestAccepts": "application/emergency.eido+json; version=\\"1,2,3\\"\"", "unitId": "urn:emergency:uid:unitid:1a12:bcaf.state.pa.us } }</pre>	The functional element returns a subscribeResponse message with a statusCode of 404 indicating that the subscription was rejected because of the unknown unit. <pre>{ "subscribeResponse": { "requestId": "63bfd612-f558-4489-b653-caa76a7edb9b", "statusCode": 404, "statusMessage": "Unit Does Not Exist" } }</pre>

3.4 MinRates and MaxRates in Detail

3.4.1 minRate

minRate refers to the minimum rate that notification messages must be sent. It is specified as a period in seconds. If a subscription notification has not been sent by the notifier within the minRate period, then the notifier must send an empty event notification message consisting of a message similar to the following example:

```
{
  "event": {
    "subscriptionId": "sub1234",
    "transactionId": "f894fjhdsh940jdf@192.168.2.1",
    "notification": []
  }
}
```

3.4.1.1 Subscriber Behavior

A subscriber that wishes to request a specific minimum notification rate MUST populate the minRate field in the subscribe message with a value greater than 0. The subscriber must be aware that the subscribe response message contains the actual minRate to be used for

the subscription. The notifier can respond with a different rate than that requested but cannot make it a shorter period than the requested rate.

3.4.1.2 **Notifier Behavior**

A notifier **MUST** consider the minRate parameter specified by the subscriber and apply local policy to determine if the specified minRate is permitted. The notifier **MAY** decide to increase the proposed minRate period based on its local policy, static configuration, or other implementation-determined constraints. The notifier **MUST** reflect the effective minimum notification period in the minRate parameter of the subscribe response message.

A notifier **MUST** generate an empty notification message when the time since the last notify message exceeds the agreed upon minRate period.

3.4.2 **maxRate**

The maxRate refers to the maximum rate that notification messages can be sent. It is specified as a period in seconds between notifications. If the effective maxRate value is 0 then maxRate will not be used.

If an event update occurs which would otherwise send a notification when the maxRate has not expired, then a notification will be generated when the maxRate period has expired. Since EIDO conveys current state, if intermediate notifications are not sent then the subscriber will receive current state when the notification is sent after the period expires.

The maxRate is applicable to all subscription types. For a subscription of type "new", when maxRate is specified and new notifications for new incidents matching the qualFilter (if supplied) exceed the maxRate, notifications are delayed until the maxRate period lapses. At that time, one notification is sent, containing one EIDO for every incident that passed the qualFilter (if specified) at any time during that period, and containing the current state of that incident. The incident state may be closed if the incident was opened and closed before the maxRate timer elapsed. In this case, maxRate delays the notifications and batches them but does not eliminate them.

For each subscription of type "single", when the maxRate period lapses and the incident changed state, a notification is sent. The EIDO contains the current state of the incident at the time the notification is sent. This means that intermediate state changes are not sent and maxRate may reduce the number of EIDOs sent to the subscriber. An EIDO specifying

that an Incident is closed is not subject to the maxRate timer. Upon Incident close, the EIDO is immediately sent after which the subscription will be terminated.

3.4.2.1 **Subscriber Behavior**

A subscriber that wishes to request a specific maximum notification rate **MUST** populate the maxRate field in the subscribe message with a value greater than 0. The subscriber must be aware that the subscribe response message contains the actual maxRate to be used for the subscription.

3.4.2.2 **Notifier Behavior**

A notifier **MUST** consider the received maxRate parameter from the subscribe message and apply local policy to determine if the received maxRate is permitted. The notifier **MAY** decide to increase the proposed maxRate period based on its local policy, static configuration, or other implementation-determined constraints. The notifier **MUST** specify the effective maximum rate of notifications in the maxRate parameter of the subscribe response message.

A notifier **MUST NOT** generate a notification if the period since the most recent notification is less than the value of the maxRate parameter. Upon establish of a subscription or renewal thereof, a notification will be sent if required and the timer associated with the maxRate will be reset.

3.5 **Filters**

Filters in Conveyance allow a subscription on a specific incident to control when EIDOs are sent. A qualification filter control notification based on the values in the EIDO that the notifier is prepared to send. The Subscribe Request **MAY** contain an optional filter component called a "qualFilter".

3.5.1 **Qualification Filter**

A qualification filter (qualFilter) specifies the criteria to be applied to an incident for a subscriber to be notified of that incident by being sent an EIDO for that incident. Example criteria include common priorities, geospatial boundaries, common incident types, and common incident statuses, as defined by the EIDO Standard.

The qualFilter is the primary mechanism to control which incidents are sent to the subscribed for a given subscription. The state of an incident can change such that an incident that did not originally pass the qualFilter may pass subsequently as the state of an incident changes. For example, the location of the incident may change which would trigger evaluation of the qualFilter. The notifier is **NOT** required to dereference location-by-reference that is mentioned in a qualFilter every time the filter is evaluated. If the notifier

does not dereference the location, it SHALL use the last location it obtained for its own purposes when evaluating the filter.

The qualFilter is relatively expensive for a notifier to compute. The more complex the qualFilter, the more compute load is required for its evaluation. Rerunning the qualFilter for each subscriber for each change in the state of an incident may require more resources than a notifier has available for that purpose. To mitigate that, an evaluation rate MAY be negotiated between the subscriber and the notifier. The subscriber specifies the qualFilterEvalPeriod, which is the integer number of seconds between per-incident evaluations requested by the subscriber. The notifier's response to the subscription request SHALL specify the qualFilterEvalPeriod which MUST be greater than or equal to the value in the subscription request. The notifier attempts to rerun the qualFilter on an incident it is considering for an EIDO notification at the negotiated rate and sends one EIDO for an incident for which:

- a) the qualFilter passes, and
- b) it has not previously sent an EIDO to the subscriber for that incident for any reason.

The notifier MAY evaluate the filter more frequently than the negotiated period if the necessary resources are available.

If the notifier's workload increases after the qualFilterEvalPeriod is negotiated, the notifier may not be able to keep up with the negotiated rate. It is RECOMMENDED that the notifier keep the maximum subscription duration in the range of 15-30 minutes, which would allow it to negotiate a new qualFilterEvalPeriod reasonably promptly in such a circumstance. Subscribers should not expect the qualFilterEvalPeriod specified in the response to remain constant from subscription to subscription for this reason.

In addition, notifiers MUST implement a scheduling mechanism. Notifiers MUST track the qualFilter compute time for incidents for each subscription containing a qualFilter. A Policy (qualFilter) provides a weight factor as a real number with a default value of 1.0. The notifier SHALL have a provisioned value that specifies the percentage of the total CPU utilization that the cumulative qualFilter evaluations for an incident is allowed to use. Each subscription SHALL be allowed to consume a weighted fraction of the allowed time. If all weights were 1.0 and all negotiated qualFilterEvalPeriod values were the same or no subscriptions have a qualFilterEvalPeriod, all qualFilter instances would use the same compute time and simpler filters would always run more often than complex filters on a saturated system.

The weight factors allow management to effect an increase or decrease in the rate a qualFilter will be run. This can be accomplished by ordering the subscriptions by the average compute time for each subscription multiplied by its qualFilter weight factor with those subscriptions whose qualFilterEvalPeriod has not elapsed placed at the end ordered

by the time of last evaluation and evaluating the incident for each subscription in that order until such time that the provisioned limit is exhausted. Any subscription not evaluated will have its compute time for that evaluation set to zero which cause the average to be lowered for that subscription. Other algorithms that achieve the same result are acceptable.

The notifier is NOT required to dereference location-by-value that is mentioned in a qualFilter every time the filter is evaluated. Rather, it uses the last location it obtained for its own purposes when evaluating the filter.

The recommended rate for qualFilterEvalPeriod is 60 seconds. Since the qualFilter may take substantial compute resource, if the notifier becomes busy after the qualFilterEvalPeriod is negotiated, it may not be able to keep up with the negotiated rate. It is RECOMMENDED that the notifier keep the maximum subscription duration in the range of 15-30 minutes, which would allow it to renegotiate the qualFilterEvalPeriod reasonably promptly in such a circumstance. Subscribers should not expect the qualFilterEvalPeriod specified in the response to remain constant from subscription to subscription for this reason.

The qualFilter criteria SHALL be specified as Basic Regular Expressions as specified in POSIX Extended Regular Expressions as defined in section 9 of IEEE Std 1003.1-2017. Data elements in the EIDO SHALL be testable variables within the expression using JSONPath [22] notation. Geospatial boundaries are described with GML [15] shapes limited to those described in OGC 06-142r1 [4]. Coordinate references MUST be EPSG 4326.

A qualFilter is a m4 [20] macro expanding to a POSIX expression where the “^” (caret) character is used as a location/shape overlap operator.

When searching text elements, partial string matches and whole “word” matches MUST be possible.

Literal strings in text elements SHALL be compared in a case-insensitive manner. A literal string in the qualFilter MUST be upper case.

Note: The POSIX expr standard does not provide for case-insensitive comparisons.

As a common extension to the POSIX standard, whole word matches SHALL be accommodated in regular expressions by surrounding the search word with “\<” and “\>”, for example, “\<word\>”.

3.5.1.1 Applying a qualFilter

The following operations are performed on the filter value:

The first operation expands the filter value using the m4 processor. The effect of the m4 syscmd macro MUST be disabled to address the security risks associated with its use. This can be accomplished by setting the SHELL environment variable to /bin/false, performing a

chroot and running the process as other than root, or equivalent. Given that m4 may recurse indefinitely, a limitation MUST be enforced to prevent such occurrence. Such a limitation may be effected by enforcing a time limit such as through the use of the POSIX timeout command or a memory limit.

The second operation replaces all instances of JSONPath references with a numeric, string, or location value obtained by locating the element in the EIDO pointed to by the JSONPath. Elements of an EIDO may contain references to other elements in the EIDO which must be dereferenced. A JSONPath reference is surrounded by a pair of bracket characters ("[" and "]"). In the case of an EIDO element that is a Boolean (i.e., "true" or "false"), the value will be zero or one. In the case of an EIDO element that is a string, the value shall be upper-cased. In the case of a geospatial reference (see below), the value will be an XML shape enclosed in braces ("{" and "}"). There is no ability to reference elements of XML documents other than PIDF-LOs [4][3] contained within JSON elements at this time.

The third operation compares each pair of geospatial references (see below) separated by the ^ (caret) character denoting the operator with 0 or 1 depending on whether the shape on the left side of the operator partially overlaps or contacts the shape on the right side.

The fourth operation submits the resulting value to the POSIX "expr" function. The expr exit value MUST be an integer. If the exit value is 0, the EIDO passes the filter. If it is non-zero, it fails.

If a geospatial reference is an EIDO element, then it must be a PIDF-LO. Otherwise, the geospatial reference must be literal GML polygon object or reference thereto. If the location object contains solely a civic address and does not contain geospatial coordinates, then the Geocode Service (GCS) as described in NENA-STA-010.3 SHALL be used to derive the geospatial coordinates that correspond to the civic address.

Geospatial references MUST use WGS84.

Geospatial references are based on The OpenGIS® Geography Markup Language Encoding Standard (GML). The Geography Markup Language (GML) is an XML grammar for expressing geographical features.

3.5.1.2 Example

The following example reflects an implementation of the qualFilter whose requirements are specified in Appendix A.

```
define('River_Location', `<?xml version="1.0" encoding="utf-8"?>
  <Polygon xmlns="http://www.opengis.net/gml">
    <exterior>
      <LinearRing>
        <posList>
          -122.30836276702803 45.558679896560022
```

```
-122.32490527625748 45.559079298970516
. . .
-122.30731697599759 45.539768832065988
-122.30836276702803 45.558679896560022
</posList>
</LinearRing>
</exterior>
</Polygon>')dnl
define(`I5_Bridge_Location', `<?xml version="1.0" encoding="utf-8"?>
<Polygon xmlns="http://www.opengis.net/gml">
<exterior>
<LinearRing>
<posList>
-122.67273541470757 45.621797702801814
-122.67318700567574 45.621947310420069
. . .
-122.67864567106085 45.611714505819862
-122.67273541470757 45.621797702801814
</posList>
</LinearRing>
</exterior>
</Polygon>')dnl
define(`I205_Bridge_Location', `<?xml version="1.0" encoding="utf-8"?>
<Polygon xmlns="http://www.opengis.net/gml">
<exterior>
<LinearRing>
<posList>
-122.55193944604855 45.60377701030086
-122.55117887140773 45.599440140350225
. . .
-122.5511218283097 45.603550863181852
-122.55193944604855 45.60377701030086
</posList>
</LinearRing>
</exterior>
</Polygon>')dnl
define(`River_Incident_Type', `.*\<(WATER|RESCUE|MISBOATR|MCI|MAYDAY|FMARIN
|EXPLOSION|DROWN|DEATH|BRGCLPSE)\>.*')dnl
define(`Bridge_Incident_Type', `.*\<(TSTOP|VEHABND|THAZ|PURSUIT|MCI|FVEH|EX
PLOSION|DISVEH|DEATH)\>.*')dnl
define(`River_Note_Keyword', `.*\<(RIVER|DROWN|DROWNING|BOAT)\>.*')dnl
define(`Bridge_Note_Keyword', `.*\<(CRASH|WRECK|ACCIDENT|FIRE)\>.*')dnl
(
(
[Incident.Location] ^ {River_Location} |
[Call.Location] ^ {River_Location\`} |
[EmergencyResource.UnitLocation] ^ {River_Location}
)
&
(
[Incident.IncidentTypeCommonRegistryText] : River_Incident_Type
```

```
|
(
[Notes.NotesAction.Comments] : River_Note_Keyword |
[AdditionalData.Notes] : River_Note_Keyword |
[Location.Notes] : River_Note_Keyword
)
)
|
(
([Incident.Location] ^ {I5_Bridge_Location}) |
([Call.Location] ^ {I5_Bridge_Location}) |
([EmergencyResource.UnitLocation] ^ {I5_Bridge_Location})
)
&
(
[Incident.IncidentTypeCommonRegistryText] : Bridge_Incident_Type
|
(
[Notes.NotesAction.Comments] : Bridge_Note_Keyword |
[AdditionalData.Notes] : Bridge_Note_Keyword |
[Location.Notes] : Bridge_Note_Keyword
)
)
|
(
(
[Incident.Location] & {I205_Bridge_Location} |
[Call.Location] & {I205_Bridge_Location} |
[EmergencyResource.UnitLocation] & {I205_Bridge_Location}
)
&
(
[Incident.IncidentTypeCommonRegistryText] : Bridge_Incident_Type
|
(
[Notes.NotesAction.Comments] : Bridge_Note_Keyword |
[AdditionalData.Notes] : Bridge_Note_Keyword |
[Location.Notes] : Bridge_Note_Keyword
)
)
)
)
```

3.5.2 Examples

client subscription request filter example:

```
{
  "subscribe":{
    "requestId":"63bfd612-f558-4489-b653-caa76a7edb9b",
    "requestType":"EIDO",
```

```
    "requestSubType": "new",  
    "requestAccepts": "application/ emergency.eido+json;  
version=\"1,2,3\"",  
    "qualFilter": "<filter_string>",  
    "minRate": 1500  
  }  
}
```

client reset expire timer example:

```
{  
  "subscribe": {  
    "requestId": "63bfd612-f558-4489-b653-caa76a7edb9b",  
    "subscriptionId": "sub1234",  
    "minRate": 1500  
  }  
}
```

client subscription request for an incidentId example:

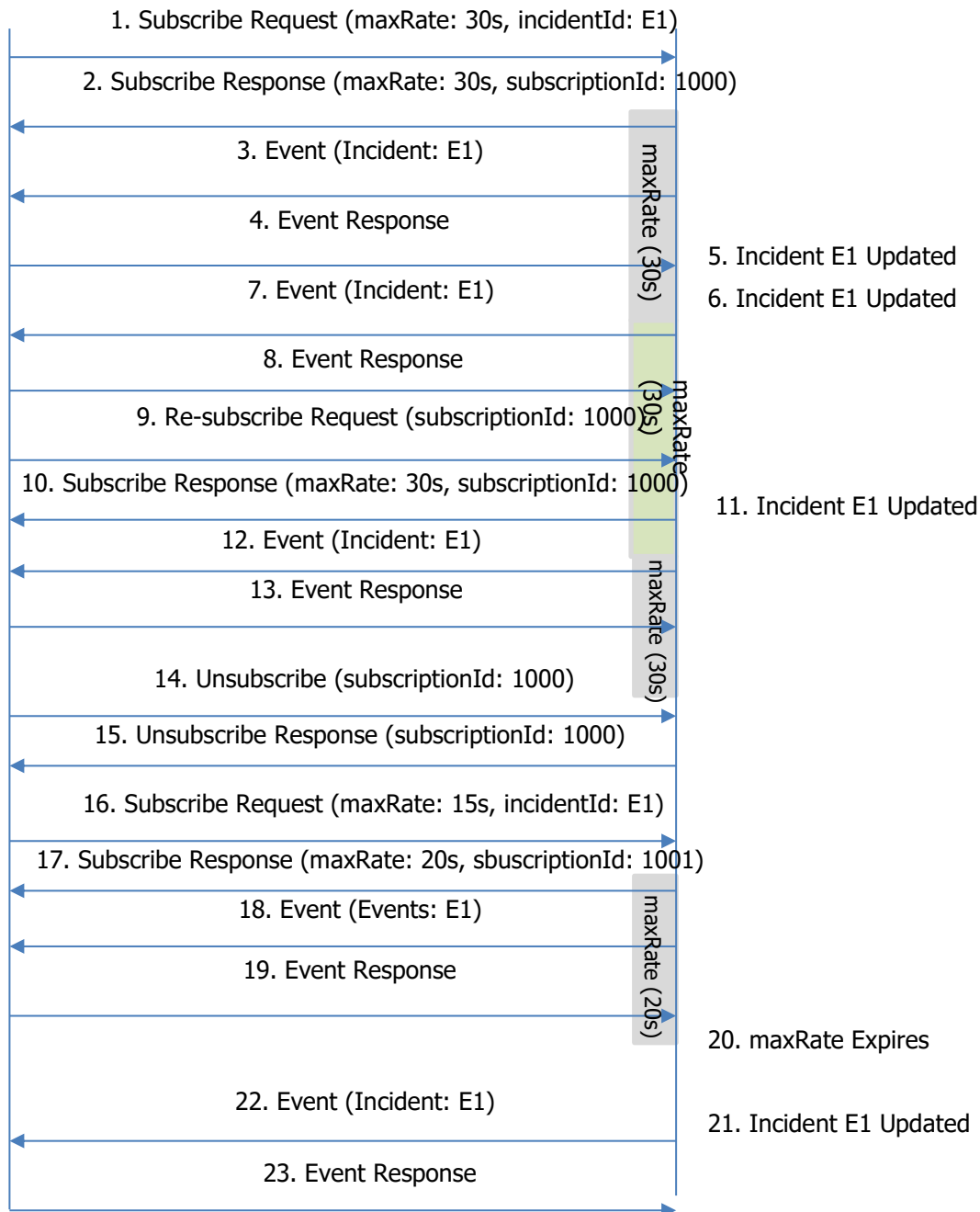
```
{  
  "subscribe": {  
    "requestId": "63bfd612-f558-4489-b653-caa76a7edb9b",  
    "requestType": "EIDO",  
    "requestSubType": "single",  
    "requestAccepts": "application/ emergency.eido+json;  
version=\"1,2,3\"",  
    "incidentId":  
    "urn:emergency:uid:incidentid:a56e556d871:bcf.state.pa.us"  
  }  
}
```

Note – The request may require an identifier related to a discrete object. For EIDO, incidentId is used to subscribe to a specific incident ID. For any other transport use case it is an event level identifier as defined by that specific protocol standard, e.g., Service or Element State Event notification it can be the SIP Call-ID of the SUBSCRIBE/NOTIFY dialog.

Subscription maxRate Example

Subscriber

Notifier



1. The subscriber sends a subscribe request to the notifier requesting to receive updates about incident E1 at a maxRate interval of 30 seconds.
2. The notifier sends a subscribe response message back to the subscriber. The notifier sets the maxRate attribute to 30 seconds to confirm that a maxRate interval of 30 seconds will be used in the conversation. The subscription is given an ID of 1000.
3. The notifier immediately sends an event message to the subscriber with the latest incident information for incident E1.
4. The subscriber responds with an event response message indicating that it successfully received the event message.
5. Incident E1 is updated. An event message is not generated since the maxRate interval has not expired.
6. Incident E1 is updated. An event message is not generated since the maxRate interval has not expired.
7. The maxRate interval elapses. The notifier sends an event message to the subscriber with the most recent EIDO state for incident E1 (i.e., The contents of the EIDO object for incident E1 will be the incident state after the updates that occurred in Steps 5 and 6).
8. The subscriber responds with an event response message indicating that it successfully received the event message.
9. The subscriber sends a re-subscribe request.
10. The notifier sends a subscribe response message back to the subscriber. The maxRate stays the same as negotiated in the initial subscribe request. The notifier does not reset the maxRate timer. (Note: a re-subscribe request maintains the subscription beyond the agreed duration of the subscription)
11. Incident E1 is updated. An event message is not generated since the maxRate interval has not expired.
12. The maxRate interval elapses. The notifier sends an event message to the subscriber with the most recent EIDO state for incident E1.
13. The subscriber responds with an event response message indicating that it successfully received the event message.
14. The subscriber would like to change the interval associated to the maxRate timer. The subscriber sends an unsubscribe message to the notifier.
15. The notifier sends an unsubscribe response message back to the subscriber indicating that the subscription has been cancelled.
16. The subscriber sends a subscribe request to the notifier requesting to receive updates about incident E1 at a maxRate interval of 15 seconds.
17. The notifier sends a subscribe response message back to the subscriber. Because of local policies the notifier sets the maxRate attribute to 20 seconds. This is the maxRate interval that will be used during the conversation. The subscription ID is now 1001.

18. The notifier immediately sends an event message to the subscriber with the latest incident information for incident E1.
19. The subscriber responds with an event response message indicating that it successfully received the event message.
20. The maxRate timer expires. A notify message is not generated because incident E1 has not been updated.
21. Incident E1 is updated.
22. Since the maxRate timer has already expired the update of incident E1 automatically generates a notify event that is sent to the subscriber.
23. The subscriber responds with an event response message indicating that it successfully received the event message.

3.6 Transport in Call Signaling

An EIDO MAY be transported in the SIP call signaling, especially when transferring an emergency call as described in the section on bridging in NENA-STA-010.

3.7 Resource Discovery

A subscriber may send a request to a notifier to request a list of resources associated with an agency. The notifier shall respond with a list of resources with their associated resource types. The request object is called resourceDiscovery and its associated response object is called resourceDiscoveryResponse.

3.7.1 Resource Discovery Request

Name	Type	Required	Description
requestId	string	Y	An ID generated by the subscriber that will be returned by the notifier in its response to the request.

Name	Type	Required	Description
requestAccepts	string	Y	The MIME type associated with the resource discovery request, which is application/emergency.eido+json, and the major versions thereof accepted as the response as a quoted comma-delimited list, as in the Accept HTTP header per RFC 7231 [11], e.g., application/emergency.eido+json; version="1" or application/emergency.eido+json; version="1,2,3". The major version is required and is the only extension parameter permitted. Any minor version associated with the major version MAY be sent.

Example:

```
{
  "resourceDiscovery": {
    "requestId": "63bfd612-f558-4489-b653-caa76a7edb9b"
    "requestAccepts": "application/emergency.eido+json;
version=\"1,2,3\"",
  }
}
```

3.7.2 Resource Discovery Response

Name	Type	Required	Description
requestId	string	Y	The request ID generated by the subscriber.
eido	object	Y	An EIDO containing the resources and associated data.
statusCode	integer	Y	A three-digit code designating the result of the request.
statusText	string	Y	A description associated with the "statusCode".

One of the following statusCode values MUST be returned.

Code	Description	Notifier Action
200	OK	None
400	Bad Request	None
403	Forbidden	None
500	Internal Server error	Retry
501	Not implemented	Retry
503	Service Unavailable	Retry
504	Gateway Timeout	Retry

Example:

```
{
  "resourceDiscoveryResponse": {
    "requestId": "63bfd612-f558-4489-b653-caa76a7edb9b",
    "eido": <EIDO>,
    "statusCode": "200",
    "statusText": "OK"
  }
}
```

3.8 Transport by Push

In some interactions, an EIDO is sent to an agency on an unsolicited basis. While the receiving entity has a policy that governs from whom it will accept an EIDO, it does not maintain a Subscription to the source.

3.8.1 Push Commands

This document defines five commands, RequestForService, ResourceMessage, RequestAwareness (collectively Request), CancelRequest, and ResponseToRequest.

A RequestForService is a request to a responder agency to request assistance with an incident, where the sending agency does not need to specify exactly what is needed, such as for the following use cases:

- A call handling FE sends a request to a dispatch FE to dispatch resources to an incident based on the dispatch FE's policies.

- Primary PSAP to a secondary PSAP where the secondary PSAP dispatches resources to an incident based on the secondary PSAP's policies.
- Primary PSAP to an out of area primary PSAP where the out of area primary PSAP dispatches resources to an incident based on out of area primary PSAP's policies.

A ResourceMessage command is used when the sending agency needs to identify a specific resource, inquire about availability of a resource, or manage dispatched resources.

A RequestAwareness is used when no incident-level response or action by the recipient is requested. If, subsequent to the RequestAwareness, action is desired, another Push MUST be made with either a RequestForService or ResourceMessage command regardless of whether or not the recipient of the RequestAwareness established a subscription to the incident. The RequestAwareness command has a required parameter named "senderPriority" whose value is an integer from 0 to 10 where 10 is most important and 0 is least important. The intent of the priority is to guide action by the recipient. Higher priority may involve more human interaction where lower priority may involve more automation.

A CancelRequest is used when a Request has previously been sent but is no longer needed.

Following the establishment of the WebSocket, the client sends messages to the FE using WebSocket text frames, which are encoded using UTF-8. The methods defined herein are requestForService, resourceMessage, requestAwareness, cancelRequest, and responseToRequest.

Payload Object	Description
{"requestForService":{<RequestForService_object>}}	Request assistance for an incident from a responding agency according to the responding agency's policies.
{"resourceMessage":{<ResourceMessage_object>}}	Request assistance for an incident from a responding agency according to the requesting agency's policies using Oasis EDXL.

Payload Object	Description
{"requestAwareness":{<RequestAwareness_object>}}	Request that an agency be aware of the incident.
{"cancelRequest":{<CancelRequest_object>}}	Cancel a Request previously made.
{"responseToRequest":{<ResponseToRequest_object>}}	Response to the Request

3.8.1.1 RequestForService Object

Name	Type	Required	Description
requestId	string	Y	An ID generated by the requester that will be returned by the responder in its response to the request.
eido	object	Y	An EIDO representing the incident for which service is requested.

3.8.1.2 RequestForService Example

```
{
  "requestForService": {
    "requestId": "63bfd612-f558-4489-b653-caa76a7edb9b",
    "eido": <EIDO>
  }
}
```

3.8.1.3 ResourceMessage Object

Name	Type	Required	Description
requestId	string	Y	An ID generated by the requester that will be returned by the responder in its response to the request.
eido	object	C	An EIDO representing the incident for which resources are requested.

Name	Type	Required	Description
resources	array	Y	An array of strings containing EDXL-RM documents referencing the resources requested.

The eido element is REQUIRED for the first request.

The ResourceMessage is simply a "wrapper" around the EDXL-RS/EDXL-RM protocol/standard messages. The EDXL processing can be viewed as operating at a higher layer in the messaging stack and therefore both EDXL resource request and response messages are transmitted using the ResourceMessage message.

When viewing it that way, the response to a ResourceMessage will only ever be a machine response indicating that the message was actually received, because what we consider the final response to the EDXL resource request will be another ResourceMessage with the EDXL response contained within it.

3.8.1.4 ResourceMessage Example

```
{
  "resourceMessage": {
    "requestId": "63bfd612-f558-4489-b653-caa76a7edb9b",
    "eido": <EIDO>,
    "resources": [<EDXL-RM>]
  }
}
```

3.8.1.5 RequestAwareness Object

Name	Type	Req'd	Description
requestId	string	Y	An ID generated by the requester that will be returned by the responder in its response to the request.
senderPriority	integer	Y	Priority to guide action by the recipient.
eido	object	Y	An EIDO representing the incident for which awareness is requested.

3.8.1.6 RequestAwareness Example

```
{
  "requestAwareness": {
    "requestId": "63bfd612-f558-4489-b653-caa76a7edb9b",
    "senderPriority": 0,
    "eido": <EIDO>
  }
}
```

3.8.1.7 CancelRequest Object

Name	Type	Required	Description
requestId	string	Y	The ID associated with the request to be canceled.

3.8.1.8 CancelRequest Example

```
{
  "cancelRequest": {
    "requestId": "63bfd612-f558-4489-b653-caa76a7edb9b",
  }
}
```

3.8.1.9 ResponseToRequest Object

Name	Type	Required	Description
requestId	string	Y	The ID provided by the requester.
responseCode	integer	Y	The responseCode element is a 3-digit integer result code indicating the response to the original request.
responseText	string	Y	The responseText provides a short textual description related to the responseCode. The value is at the discretion of the implementor.

3.8.1.10 ResponseToRequest Example

```
{
  "response": {
    "requestId": "63bfd612-f558-4489-b653-caa76a7edb9b",
    "responseCode": 501,
    "responseText": "Not implemented"
  }
}
```

3.8.1.11 Notes

For a given recipient, a RequestAwareness is automatically canceled and superseded by the receipt of either a RequestForService or a ResourceMessage.

For RequestForService and RequestAwareness, the EIDO, which MUST include an incident data component, is included in the request.

For ResourceMessage, the EIDO, which MUST include an incident data component, is included in the first request for a given EDXL OriginatingMessageID.

A recipient that responds to the Request with a 200 (OK) response MUST create an active incident within that agency based on the EIDO that accompanied the Request and each of the requestor and the recipient MAY subscribe to the other's incident. The recipient MAY close the incident immediately, which changes the state of the incident at that agency, in which case there is no need for the recipient to subscribe to the requestor's incident.

3.8.1.12 ResponseToRequest Status Codes RequestForService

The following status codes have the following meanings:

Code	Description	Notes
180	Offered	The Request has been presented to one or more agents
182	Queued	The Request has been queued for presentation to an agent
200	OK	The Request has been accepted
302	Moved Temporarily	The Request should be resubmitted to the location specified
403	Forbidden	Request refused due to policy non-compliant
404	Not Found	Recipient does not provide for the Request
408	Request Timeout	None

Code	Description	Notes
488	Not Acceptable Here	Requested dispatch type/resources are not provided by the agency and Reason will specify why
500	Server Internal Error	Find an alternate agency
501	Not Implemented	Find an alternate agency
502	Bad Gateway	Find an alternate agency
503	Service Unavailable	Find an alternate agency
504	Server Time-out	Either retry or find an alternate agency based on policy
505	Version Not Supported	Find an alternate agency
513	Message Too Large	This may be returned if the EIDO exceeds recipient's maximum which must be greater than or equal to ten mebibytes. Reduce the size of the EIDO and try again or find an alternate agency.
603	Decline	Request cannot be fulfilled
606	Not Acceptable	Requested agency does not provide service in requested area

RequestAwareness

The following status codes have the following meanings:

Code	Description	Notes
200	OK	Request accepted for consideration and an incident has been created to which the requester can subscribe
302	Moved Temporarily	The Request should be resubmitted to the location specified
403	Forbidden	Request refused due to policy non-compliant
404	Not Found	Recipient does not provide for the Request
408	Request Timeout	
486	Busy Here	System resources are occupied for this Request; Retry
488	Not Acceptable Here	No action will be taken by the recipient. Reason header may specify why.
500	Server Internal Error	
501	Not Implemented	
502	Bad Gateway	

Code	Description	Notes
503	Service Unavailable	
504	Server Time-out	Retry
505	Version Not Supported	
513	Message Too Large	This may be returned if the EIDO exceeds the recipient's maximum which must be greater than or equal to ten mebibytes.

RequestAwareness, ResourceMessage, and RequestForService Status Codes

If a recipient returns a 200 status code, the sender should immediately follow that with a subscription (see section 3.3.4.1) to receive updates to the recipient's incident.

Recipient SHALL return an EDXL-RM solely in response to a ResourceMessage.

ResourceMessage and CancelRequest

The following status codes have the following meanings:

Code	Description	Subscriber Action
200	OK	None
400	Bad Request	None
408	Request Timeout	None
413	Payload too large	None
481	Subscription does not exist	None
500	Internal Server Error	None
503	Service Unavailable	Retry request
513	Message Too Large	This may be returned if the EIDO exceeds the recipient's maximum which must be greater than or equal to ten mebibytes.

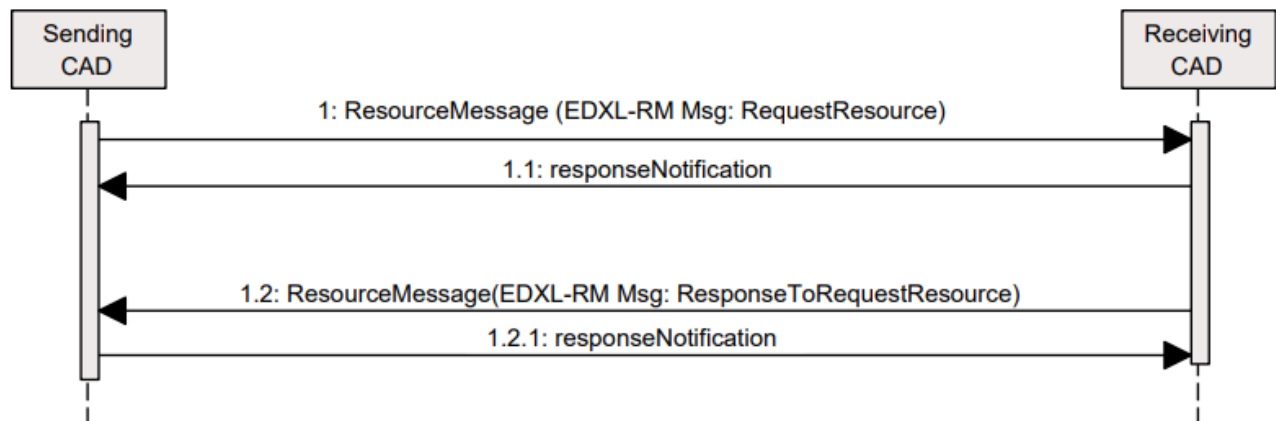
The CancelRequest is expected to be handled by the recipient with no human intervention so there are no intermediate status codes.

3.8.2 Requesting Specific Resources

Where the sending agency needs to identify a specific resource, inquire about availability of a resource, or manage dispatched resources such resource requests use the OASIS EDXL-RM 1.0 resource management mechanism [16]. The specification for EDXL-RM uses the

EDXL Distribution Element [17], however, the Distribution Element is not used in EIDO Conveyance. The EDXL-RM message MUST be included in addition to the EIDO.

The request for resources is a series of ResourceMessage messages between the sending and receiving agencies to manage resource requests. The actual negotiation for the resources is managed by the EDXL-RM standard, and ResourceMessage facilitates that negotiation. The behavior of the sending and receiving agencies MUST implement the behavior documented in the EDXL-RM standard so that the resource negotiation occurs correctly. An example of a simple negotiation is illustrated below.



3.8.2.1 Messages Applicable to EIDOs

EDXL-RM defines a large number of messages. All of them could be seen in an EIDO “push” operation, although the most common “push” EIDO exchanges will only use the “RequestResource” and “ResponseToRequestResource” messages. The following sections describe the use of the EDXL-RM message set:

RequestResource: used to actually request action, or to ask if resources needed would be available as follows. When sent with a generic resource request, this message is what would be used by a PSAP requesting a responder to dispatch appropriate resources. When more specific resources are needed, this recipient can agree to send them in response to this request, or it can indicate they are available. In the latter case, the RequisitionResource message is used to actually request the available resources.

ResponseToRequestResource: a response request is always responded to by the agency to whom the RequestResource was sent. The Response indicates whether the agency receiving the request will respond to the request, specify resources that could be available, or decline the request-

RequisitionResource: EDXL-RM allows the RequestResource message to be used as a form of generalized “could you help?” request, which does not result in commitment of

resources but rather the availability thereof. In such cases, RequisitionResource is used to request actual action be taken.

CommitResource: in response to a RequestResource, or a RequisitionResource, a CommitResource signals the recipient that the requested resources are committed to the incident. For a generic request for action, the “accepted” status of a ResponseToRequestResource is sufficient. Where more specific resource requests are made, CommitResource indicates that resources are assigned to the incident.

RequestInformation: used to ask questions about specific resources.

ResponseToRequestInformation: response to RequestInformation.

OfferUnsolicitedResource: sometimes, an agency learns of an incident, or a resource request by some method not described in this section. Examples include “self-dispatch” to a major incident. This message must be used by agencies that wish to offer help on an incident.

ReleaseResource: when a requesting agency no longer needs a resource, it notifies the responding agency with a ReleaseResource message.

RequestReturn: if a responding agency needs resources it has allocated to an incident back, it uses RequestReturn.

ResponseToRequestReturn: used to respond to a RequestReturn message.

RequestQuote: rarely used in NG9-1-1. In some circumstances, where a resource can be provided, but only if the requesting agency agrees in advance to pay for the resource, RequestQuote is used to enquire what the cost will be.

ResponseToRequestQuote: used to respond to RequestQuote.

RequestResourceDeploymentStatus: where a resource has been committed, but is not yet on-scene, RequestResourceDeploymentStatus can be used to inquire about the resource.

ReportResourceDeploymentStatus: response to RequestResourceDeploymentStatus

RequestDeploymentExtendedDuration: when a resource is provided, but the requesting agency needs to extend the time the resource is needed, this message can be used to make that request.

ResponseToRequestDeploymentExtendedDuration: response to RequestDeploymentExtendedDuration.

3.8.2.2 EDXL Message Components

ResourceMessage commands MUST conform to the specifications in EDXL-RM with the following restrictions. Implementations SHOULD accept the full generality of the EDXL-RM specification on receipt and respect these restrictions on send.

MessageId MUST be a URN formed by the prefix "urn:emergency:uid:edxl-rm:", a unique string containing alpha and/or numeric characters, the ":" character, and the element identifier of the element that sends the message. Note that the Distribution Element also has an ID, and it is the same as this ID. Each EDXL resource message contains a MessageId that uniquely identifies the resource message.

OriginatingMessageID identifies the MessageID of the first message in a message sequence to which the message belongs. If the message is itself the originating message in a new sequence, OriginatingMessageID will have the same value as the MessageID element. In some other cases, the OriginatingMessageID element will have the same value as the PrecedingMessageID element. The OriginatingMessageID value essentially forms a unique identifier for a group of related messages, linking them together so that the relationship between the messages is made explicit and unambiguous (and threads of messages can be tracked by resource management software).

PrecedingMessageID identifies the message that immediately preceded the current message in the message sequence. This MessageID is an EDXL-RM MessageID not an EDXL-Distribution Element MessageID.

SentDateAndTime MUST be a TimeStamp (that is, a date and time formatted perNENA-STA-010).

ContactInformation MUST be a xpi:PartyType, and the values MUST be mapped to/from an xCard of the same party.

IncidentID MUST contain an IncidentId and the content MUST be an NG9-1-1 Incident ID

ReasonCode MUST contain a value from the EDXLRMReasonCode Registry

Resource SHOULD contain a Resource:TypeStructure, which SHOULD use a ValueListURN from the EDXLRMresourceLists Registry. If the resource requested is not in a list, the Resource Element MAY contain a Resource:Name. In some circumstances, such as mutual aid or automatic aid, where the sender and the recipient have made prior arrangements, a resource list not in the EDXLRMresourceList registry MAY be used. The Resource:Name nominates a type of resource, not the ID of a specific resource.

Owner MUST be the first component of the "ORG" value in the Agency xCard contained in the AgencyLocator record for the owning agency.

OwningJurisdiction MUST be the name of a state, county, incorporated municipality or unincorporated community, and SHOULD match the name of one such boundary in the ECRF. Where the jurisdiction consists of multiple such boundaries, an OwnerType SHOULD be used in preference to an OwningJurisdictionType.

HomeDispatch MUST be the first component of the "ORG" value in the Agency xCard contained in the AgencyLocator record for the owning agency.

Location MUST be an OASIS Address (a CIQ structure containing address details in an internationally-applicable format) or an OASIS TargetArea (a container element for GML-based geospatial information). If the Location is the incident location from the EIDO then mapping from the EIDO Incident Location to EDXL-RM Location MUST be performed, and, if Location is a geospatial reference, only Point (WGS84 latitude and longitude), CircleByCenterPoint (a point and radius) and Polygon (a set of connected non-crossing lines) are allowed.

3.9 EIDO Dereference Factory

The agency's IDX MAY maintain an internal EIDO RESTful web service URL that, when an HTTP GET whose last element of the path is the incident ID is presented, returns a Dereference Service URI for the Incident. Data Rights Management would control if the issuance of a URI from the Dereference Factory is permitted. The Dereference Factory MUST supply the same URI for a given requestor and incident so that the URI expiration mechanism cannot be circumvented.

For example:

```
https://psap.alleghany.pa.us/IncidentData/v1/dereference/urn:emergency:uid:  
:incident:6955@psap.alleghany.pa.us
```

HTTP request:

```
GET /  
IncidentData/v1/dereference/urn%3Aemergency%3Auid%3Aincident%3A6955%40psap  
.alleghany.pa.us HTTP/1.1  
Host: psap.alleghany.pa.us
```

Result is:

```
https://psap.alleghany.pa.us/foobar/asdasdsadasadasd
```

HTTP request:

```
GET /foo/foobar/asdasdsadasadasd HTTP/1.1  
Host: psap.alleghany.pa.us
```

Result is:

{I AM AN EIDO}

One of the following HTTP status codes and descriptions MUST be returned:

Code	Description
200	OK
403	Forbidden
404	Not found
406	Not Acceptable
408	Request Timeout
500	Internal Server Error

3.10 EIDO Dereference Service

Any entity in possession of the above-referenced EIDO URI may issue an HTTP GET on that URI and a full EIDO will be returned, although normal authentication and authorization mechanisms as described above MUST be used to limit what entities may do any form of dereference. Once used, the URI expires for the dereferencer and may not be used again by that dereferencer. The Dereference service MUST NOT provide more than one dereference for the URI from a given dereferencer. If updates to the incident are required, use the subscription mechanism.

One of the following HTTP status codes and descriptions MUST be returned:

Code	Description
200	OK
403	Forbidden
404	Not found
406	Not Acceptable
408	Request Timeout
500	Internal Server Error

3.10.1 Request and Response

The GET request MUST contain an Accept: header which specifies the MIME type assigned to EIDO (application/emergency.eido+json) and MUST include as a parameter a comma-delimited list of the major version(s) of the schema the client supports (for example, Accept: application/emergency.eido+json; version="1,2,3"). If the server can fulfil the

request, the response MUST include one and only one EIDO instance in the body of the reply and its version MUST be the highest mutually compatible major version. The client MUST expect to receive an object derived from any minor version of the specified EIDO schema, including a higher minor version than it currently supports. The client MUST ignore any fields it does not understand. If the server does not support any of the major versions found in the Accept: header of the GET request, it MUST return a 406 Not Acceptable response.

3.11 EIDO Retrieval Service

An agency's IDX MUST provide an EIDO Retrieval Service which is a RESTful web service URL that, when an HTTP GET whose last element of the path is the incident ID is presented, returns an EIDO. The GET request MUST contain an Accept: header which specifies the MIME type assigned to EIDO (application/emergency.eido+json) and MUST include as a parameter a comma-delimited list of the major version(s) of the schema the client supports (for example, Accept: application/emergency.eido+json; version="1,2,3"). If a request is received without the Accept: header, it MUST be assumed that the request is for version 1. The content of the EIDO returned is limited by Data Rights Management policy. The EIDO Retrieval Service URL for an Agency can be found in the Agency Locator record for that agency as specified in NENA-STA-010¹.

The requirements defined in section 3.10.1 above also apply to this service.

For example:

```
https://idx.esinet.net/IncidentData/V1/incidents/urn:emergency:uid:incident:6955@psap.alleghany.pa.us
```

HTTP request:

```
GET
/IncidentData/V1/incidents/urn%3Aemergency%3Auid%3Aincident%3A6955%40psap.
alleghany.pa.us HTTP/1.1
Host: psap.alleghany.pa.us
Accept: application/emergency.eido+json; version="1,2,3"
```

Result is:

```
{I AM AN EIDO}
```

One of the following HTTP status codes and descriptions MUST be returned:

Code	Description
200	OK
403	Forbidden

¹ Anticipated to be in NENA-STA-010.4.

Code	Description
404	Not found
406	Not Acceptable
408	Request Timeout
500	Internal Server Error

3.12 Data Rights Management

All FEs in an agency MUST have the same Data Rights Management (DRM) policy and therefore it is not important to know which FE created any particular data component. The only FE that can include a data component in an EIDO that it is sending, which it received in an EIDO, is the IDX. The IDX will be defined in future work, as mentioned in Section 5.4.1. An agency IDX MUST NOT send data received from another agency, except as noted below. An inter-agency IDX MAY send data received from other agencies but MUST respect the data rights management policy of the FE (typically an IDX) that sent it the data.

The EIDO Incident component contains data vital to any agency participating in an incident; therefore, any agency MAY include in an EIDO it sends, information from the Incident data component it received from another agency, subject to its own data rights management policy.

Similarly, subject to its own data rights management policy, an agency receiving an EIDO from another agency related to an incident in which three or more agencies are involved MUST include the Dispatch data component information it received that identifies all of the agencies involved.

The content of an EIDO, regardless of the way it is conveyed, MUST comply with the data rights management policy of the agency that created the data being incorporated into the EIDO, controlled by the DRM mechanism defined in NENA-STA-010. The data rights management policy MUST conform to local, state/provincial, and federal laws and regulations. If an entity receives an EIDO from another agency, and subsequently sends an EIDO to another agency, it MUST filter the EIDO it sends to comport with the EIDO policy of the agency from which it received the information with respect to the data components it is copying.

Filtering of EIDO content to conform with DRM policy SHALL be accomplished by deleting data components or elements thereof in their entirety from the EIDO being sent, based on the policy.

In the DRM policy language as described in NENA-STA-010, policy rules are expressed in an "if [condition] then [action]" statement. The action "permit" allows a data component or element thereof to be included in the EIDO, while "deny" disallows that component or element thereof from being included in the EIDO. The name of the data component or

element prefixed by "EIDO." is used to name the resource to which the rule refers, for example, "EIDO.Agent". The EIDOConveyance entry in the IANA Interface Names registry will be used to control the access rights for EIDO data components or elements thereof.

DRM is also used to control which entities are permitted to subscribe to an incident. The existing conditions provide for permit or deny based on agency name, type, etc. This document adds two new subject attributes, "pusher" and "pushee", that provide for rules to permit or deny a subscription to an incident for the agency that sent an EIDO or that received an EIDO respectively. These subject attributes may also be used to filter content.

An example is a PSAP on one side of the country pushes an EIDO to a PSAP on the other side of the country as a RequestForService. The objective is for the recipient (pushee) to permit the pusher to subscribe to the incident and for the pusher to permit the pushee to subscribe to the incident.

3.13 Discrepancy Reporting

An entity MAY report a discrepancy using the discrepancy reporting mechanism defined in NENA-STA-010. The discrepancy reports related to Conveyance are:

- A request was rejected that the requester believes should have been accepted
- The EIDO did not conform to the JSON schema
- The content of an EIDO where a value is inconsistent with the domain of values appropriate for that element
- The content of an EIDO appears inconsistent with the incident associated with the subscription
- Elements in the EIDO that were expected were absent perhaps due to permissions
- A notification was received that did not match the filter associated with the subscription
- A notification was not received that should have based on the filter associated with the subscription

When an EIDOConveyanceDiscrepancyReport is submitted, the "problemService" parameter is set to "EIDOConveyance" and the object contains the following elements:

Name	Type	Required	Description
request	string	C	The request JSON
response	string	C	The response JSON

Name	Type	Required	Description
subscriptionId	string	C	The subscription ID
problem	string	Y	The problem being reported
eventTimestamp	string	Y	Timestamp of event being reported

request, response, and subscriptionId are REQUIRED if they exist when the discrepancy concerns a subscription.

The value of problem MUST be one of the following:

- NotAccepted
- NonConformant
- BadDomain
- EIDOInconsistent
- EIDOMissingElement
- NotificationUnexpected
- NotificationExpected

The Resolution parameter in an EIDOConveyance DiscrepancyResolution report contains one of the following tokens:

- DiscrepancyCorrected
- PerPolicy (behavior was per policy, hence not a discrepancy)
- NoDiscrepancy
- OtherResponse

3.14 Logging

Regardless of transport mechanism (i.e., subscribe, message, dereference, or included in a call transfer):

Any FE that sends or receives an EIDO MUST log it to a Logger FE with as defined in NENA-STA-010 with an EIDO LogEvent [24].

Certain LogEvents are defined in NENA-STA-010. One of those, EidoLogEvent, is extended. In addition, the following LogEvent types are defined, some of which have common members: The "direction" member has one of two values, "incoming" and "outgoing". The "queryId" member is used to relate the query to the response. The value of the "queryId" member is generated locally, MUST be globally unique, and is suggested to be of the form

"urn:nena:uid:logEvent:", followed by a locally unique ID, followed by a colon, followed by the domain of the entity performing the logging, for example, "urn:nena:uid:logEvent:a99dasdas37:psap.example.com". For those requests and responses for which there is a "requestId" element, that will take the place of the "queryId" element. In most of these events, a "peerId" member is included, which value is the identity of the peer. For a client or subscriber, it is the identity of the server or notifier. For the server or notifier, it is the identity of the requesting client or subscriber.

For any transmission for which the sender does not receive a response, the failure **MUST** be logged with the applicable error or response LogEvent using status 408 (Request Timeout).

3.14.1 EidoLogEvent

The EidoLogEvent contains the following elements:

Name	Type	Required	Description
direction	string	Y	"incoming" or "outgoing"
peerId	string	C	The identity of the peer entity
reference	string	C	An EIDO reference
requestId	string	C	The ID associated with a request
subscriptionId	string	C	The ID associated with a subscription
body	object	C	An EIDO

The "peerId" member **MUST** be populated when the EidoLogEvent results from a subscription or a request.

The "reference" member **MUST** be populated when the EidoLogEvent results from a reference in which case the "body" element **SHALL NOT** be present.

The "requestId" member **MUST** be populated when associated with a request.

The "subscriptionId" member **MUST** be populated when the EidoLogEvent results from a subscription.

The "body" member **MUST** be populated when passed by value in which case the "reference" member **SHALL NOT** be present.

The "peerId" member will be required in a future version of the schema.

3.14.2 EidoDereferenceFactoryQueryLogEvent

The EidoDereferenceFactoryQueryLogEvent contains the following elements:

Name	Type	Required	Description
direction	string	Y	"incoming" or "outgoing"
peerId	string	Y	The identity of the peer entity
queryId	string	Y	The ID associated with a dereference request

In addition, the common LogEvent prolog member, "incidentId", MUST be the Incident Identifier associated with the query.

3.14.3 EidoDereferenceFactoryQueryResponseLogEvent

The EidoDereferenceFactoryQueryResponseLogEvent contains the following elements:

Name	Type	Required	Description
direction	string	Y	"incoming" or "outgoing"
peerId	string	Y	The identity of the peer entity
queryId	string	Y	The ID associated with a dereference request
generatedURI	string	C	The URI generated by the Dereference Factory that may be used to fetch the EIDO
responseError	string	C	An error code from the Error Codes Registry
responseText	string	C	A text description of the error

The EidoDereferenceFactoryQueryResponseLogEvent MUST include either:

- "generatedURI" if successful , or
- "responseError" AND "responseText" if unsuccessful.

3.14.4 EidoDeniedLogEvent

If a client has requested an EIDO from the Dereference Service and that request has been denied or deemed invalid, it MUST be logged using an EidoDeniedLogEvent by both the Dereference Service and the derefencer. The EidoDeniedLogEvent contains the following elements:

Name	Type	Required	Description
direction	string	Y	"incoming" or "outgoing"
peerId	string	Y	The identity of the peer entity
eidoDereference	string	Y	The URI that was attempted to be dereferenced
reasonCode	string	Y	A reason code from the EIDO Dereference Denial Registry
reasonText	String	Y	A description of the error

In addition, the common LogEvent prolog member, "incidentId", MUST include the Incident Identifier if known.

3.14.5 EidoTransmissionErrorLogEvent

If an error in the process of sending or receiving an EIDO occurs, the error MUST be logged using an EidoTransmissionErrorLogEvent. A successful transmission will be logged with an EidoLogEvent. The EidoTransmissionErrorLogEvent contains the following elements:

Name	Type	Required	Description
direction	string	Y	"incoming" or "outgoing"
peerId	string	Y	The identity of the peer entity
errorCode	string	Y	A reason code from the EIDO Transmission Status Registry
errorText	string	Y	A description of the error

In addition, the common LogEvent prolog member, "incidentId", MUST include the Incident Identifier.

3.14.6 SubscriptionRequestedLogEvent

The subscriber MAY and the notifier MUST log a subscription requested log event using SubscriptionRequestedLogEvent. The SubscriptionRequestedLogEvent contains the following elements:

Name	Type	Required	Description
protocol	string	N	Used to identify the message protocol, "emergency-ent", which is a value found in the subscription protocols registry.
direction	string	Y	"incoming" or "outgoing"
peerId	string	Y	The identity of the peer entity
queryId	string	Y	The ID of the request (requestId in the request)
requestType	string	C	Type of request, e.g., "EIDO".
requestSubType	string	C	The requestSubType provides for different actions based on the requestType specified.
requestAccepts	string	C	The MIME type associated with the requestType.
expires	string	C	The requested expiry time in the "expires" member
subscriptionId	string	C	The subscription ID
qualFilter	string	C	The requested qualFilter
qualFilterEvalPeriod	integer	C	If qualFilter specified, a positive integer specifying the requested number of seconds between qualFilter evaluations.
minRate	integer	C	Number of seconds after the last notification after which a notification must be sent consistent with RFC 6446 [9][9]. If not specified there is no minimum rate.
maxRate	integer	C	Minimum time in seconds to wait between notifications which is the maximum rate at which notifications are sent consistent with RFC 6446 [9]. If not specified there is no maximum rate.
unitId	string	C	The unit ID of the requested resource

For any element where "Required" is "C" (conditional), the log event MUST include that element if specified in the subscription request.

In addition, the common LogEvent prolog member, "incidentId", MUST include the Incident Identifier if specified in the subscription request.

3.14.7 SubscriptionRequestedResponseLogEvent

The SubscriptionRequestedResponseLogEvent contains the following elements:

Name	Type	Required	Description
direction	string	Y	"incoming" or "outgoing"
queryId	string	Y	The request ID
subscriptionId	string	C	The subscription ID
qualFilterEvalPeriod	integer	C	Negotiated number of seconds between qualFilter evaluations. If requested, may be a longer period than the request but not shorter.
minRate	integer	C	May be a longer period than requested but not shorter.
maxRate	integer	C	May be a shorter period than requested but not longer.
expires	integer	C	The number of seconds after which the subscription will expire relative to its establishment
errorCode	string	C	A three-digit error code
errorText	string	C	A description of the error

For any element where "Required" is "C" (conditional), the log event MUST include that element if specified in the subscription response.

3.14.8 SubscriptionTerminatedLogEvent

The SubscriptionTerminatedLogEvent contains the following elements:

Name	Type	Required	Description
peerId	string	Y	The identity of the peer entity
direction	string	Y	"incoming" or "outgoing"
queryId	string	Y	The request ID

Name	Type	Required	Description
reason	string	Y	The reason the subscription was terminated.
subscriptionId	string	Y	The ID of the subscription

3.14.9 SubscriptionTerminatedResponseLogEvent

The SubscriptionTerminatedResponseLogEvent contains the following elements:

Name	Type	Required	Description
peerId	string	Y	The identity of the peer entity
direction	string	Y	"incoming" or "outgoing"
queryId	string	Y	The query ID specified in the SubscriptionTerminatedLogEvent
subscriptionId	string	Y	The ID of the subscription
statusCode	string	Y	A three-digit code designating the result of the request
statusText	string	Y	A description associated with the "statusCode"

3.14.10 WebSocketEstablishedLogEvent

The WebSocketEstablishedLogEvent contains the following elements:

Name	Type	Required	Description
peerId	string	Y	The identity of the peer entity
direction	string	Y	"incoming" or "outgoing"
websocketId	string	Y	A globally unique ID for a WebSocket connection
statusCode	string	Y	A code describing the result of the WebSocket establishment.
statusDescription	string	Y	A text description associated with the code

The value of the “websocketId” member is generated locally and MUST be globally unique. It is suggested to be of the form “urn:nena:uid:logEvent:”, followed by a locally unique ID, followed by a colon, followed by the domain of the entity performing the logging, for example, “urn:nena:uid:logEvent:a99dasdas37:psap.example.com”.

3.14.11 **WebSocketTerminatedLogEvent**

The WebSocketTerminatedLogEvent contains the following elements:

Name	Type	Required	Description
peerId	string	Y	The identity of the peer entity
direction	string	Y	“incoming” or “outgoing”
websocketId	string	Y	A globally unique ID for a WebSocket connection
closeCode	string	Y	A WebSocket Close frame status code
closeReason	string	N	A text description the web socket closure containing reason per RFC 6455 [10]

3.14.12 **RequestForServiceLogEvent**

The requester and the responder MUST log a request for service log event. The RequestForServiceLogEvent contains the following elements:

Name	Type	Required	Description
direction	string	Y	“incoming” or “outgoing”
peerId	string	Y	The identity of the peer entity
requestId	string	Y	The ID used to match the request with the response and the associated EidoLogEvent

3.14.13 **RequestAwarenessLogEvent**

The requester MUST and the responder MAY log a request awareness log event. The RequestAwarenessLogEvent contains the following elements:

Name	Type	Required	Description
direction	string	Y	“incoming” or “outgoing”

Name	Type	Required	Description
peerId	string	Y	The identity of the peer entity
requestId	string	Y	The ID used to match the request with the response and the associated EidoLogEvent

3.14.14 **CancelRequestLogEvent**

The requester MUST and the responder MAY log a cancel request log event. The CancelRequestLogEvent contains the following elements:

Name	Type	Required	Description
direction	string	Y	"incoming" or "outgoing"
peerId	string	Y	The identity of the peer entity
requestId	string	Y	The ID of the cancellation request used to match the request with the response and the associated EidoLogEvent

3.14.15 **ResponseToRequestLogEvent**

The requester MAY and the responder MUST log a response to request response log event. The ResponseToRequestLogEvent contains the following elements:

Name	Type	Required	Description
direction	string	Y	"incoming" or "outgoing"
peerId	string	Y	The identity of the peer entity
responseCode	string	Y	A three-digit code identifying the type of response
responseText	String	Y	A description associated with the response code

3.14.16 **ResourceMessageLogEvent**

The ResourceMessageLogEvent contains the following elements:

Name	Type	Required	Description
direction	string	Y	"incoming" or "outgoing"

Name	Type	Required	Description
peerId	string	Y	The identity of the peer entity
resourceMessage	object	Y	A ResourceMessage object

3.14.17 ResourceDiscoveryLogEvent

The ResourceDiscoveryLogEvent contains the following elements:

Name	Type	Required	Description
direction	string	Y	"incoming" or "outgoing"
peerId	string	Y	The identity of the peer entity
emergencyResourceComponent	object	Y	An emergencyResourceComponent object

3.15 Security

All transports specified in this document are protected by TLS using credentials (for those entities within an ESInet) traceable to the PSAP Credentialing Agency (PCA).

Data rights management as specified above limits the information that may be included in any given EIDO.

4 IANA Actions

4.1 WebSocket Subprotocol Name Registry

IANA is requested to add the following values to the WebSocket Subprotocol Name Registry.

Subprotocol Identifier	Subprotocol Common Name	Subprotocol Definition
emergency-ent1.1	Emergency Event Notification Transport 1.1	<this document>

4.2 Subscription Protocol Name Registry

Processing of emergency calls uses a variety of services which are queried either using SIP or HTTP or WebSocket. The logging of the events related thereto require recording the protocol used and codify their names in a registry. IANA is requested to create the subscriptionProtocolName registry in the Emergency Area.

4.2.1 **Name**

subscriptionProtocolName

4.2.2 **Information Needed to Create a New Value**

Identifier and common name of the subscription protocol and a reference to a document that specifies the subscription protocol.

4.2.3 **Registration policy**

Specification Required [RFC8126], which requires expert review. The expert should confirm that entry represents a new type of service.

4.2.4 **Content**

This registry contains:

An ASCII "Protocol Identifier" of the protocol (e.g., "sip")

An ASCII "Protocol Common Name" of the protocol (e.g., "Session Initiation Protocol")

A "Protocol Definition", a URI to the document that defines the subscription protocol.

Protocol Identifier	Protocol Common Name	Protocol Definition
sip	Session Initiation Protocol	<this document>
emergency-ent	Emergency Event Notification Transport	<this document>

5 **Impacts and Considerations**

5.1 **Operations Impacts Summary**

Inter-agency agreements and standard operating procedures will need to be established or revised to address the operational impact of the exchange of EIDOs between agencies. Agencies will be required to analyze their needs with respect to the implementation of the publishing and subscribing of EIDOs.

5.2 **Technical Impacts Summary**

The exchange of EIDOs between functional elements introduces a significant increase in the complexity of interoperable sharing of data relating to incidents. Introducing standardized data consisting of standardized elements inside an agency or between agencies fundamentally affects design and interoperability of components. Subsystems and interfaces within agencies will need to accommodate these specifications.

Network connectivity must be established between parties. In some circumstances, large volumes of EIDOs may be exchanged and the network and logging capacity to handle the volume of data must be planned for. Monitoring of performance will be required.

5.3 Security Impacts Summary

Credentials need to be provisioned in agencies and functional elements exchanging EIDOs.

Agencies will need to create data rights management policies to determine what can be conveyed in EIDOs.

Some entities sending or receiving EIDOs may be outside the ESInet, e.g., tow truck operators. Such entities may need special consideration to ensure appropriate security measures are in place, e.g., firewalls, etc.

5.4 Recommendation for Additional Development Work

5.4.1 Incident Data eXchange (IDX)

This standard makes references to the IDX which is defined as a separate effort. The i3 Architecture Working Group is anticipating release of IDX text in STA-010.3.1.

5.4.2 Signatures

Consideration should be given to non-repudiation of sending and receiving via signing and the negotiation of the requirement thereof and encryption related thereto based on in STA-010.3.0.

5.4.3 Content Filters

Consideration should be given to adding support for Content Filters based on RFC 4661 [8].

5.4.4 Location Filters

Consideration should be given to adding support for Location Filters based on RFC 6447 [27].

5.4.5 Resource Discovery Filters

Consideration should be given to adding support for Resource Discovery Filters, such as, emergencyResourceTypeCommonRegistryText, resourceAttributeRegistryText, primaryUnitStatusRegistryText, and secondaryUnitStatusRegistryText.

5.5 Anticipated Timeline

Implementation and deployment will take place as required.

5.6 Cost Factors

This standard should result in a long-term substantial reduction in the cost of new interfaces that exchange incident data between functional elements of multiple vendors.

5.7 Cost Recovery Considerations

Not applicable.

5.8 Additional Impacts (non-cost related)

None known at this time.

6 Abbreviations, Terms, and Definitions

See the NENA Knowledge Base Glossary at <https://kb.nena.org> for a complete listing of terms used in NENA documents. All abbreviations used in this document are listed below, along with any new or updated terms and definitions.

Term or Abbreviation (Expansion)	Definition / Description
EDXL-RM (Emergency Data Exchange Language – Resource Messaging)	An OASIS standard that describes a suite of standard messages for sharing data among information systems that coordinate requests for emergency equipment, supplies, and people.
EIDO (Emergency Incident Data Object)	A JSON-based (JavaScript Object Notation) object that is used to share emergency incident information between and among authorized entities and systems. NENA has adopted the JSON-based EIDO (Emergency Incident Data Object) for sharing incident information among authorized NG9-1-1 entities and systems.

Term or Abbreviation (Expansion)	Definition / Description
expr	expr is a command line utility on Unix and Unix-like operating systems which evaluates an expression and outputs the corresponding value [19]
IDX (Incident Data eXchange)	A Functional Element that facilitates the exchange of Emergency Incident Data Objects (EIDOs) among other Functional Elements both within and external to an agency.
IEEE (Institute of Electrical and Electronics Engineers)	A professional association that develops, defines, and reviews electronics and computer science standards.
IETF (Internet Engineering Task Force)	Lead standard-setting authority for Internet protocols.
JSON (JavaScript Object Notation)	A lightweight data-interchange format based on a subset of the JavaScript Programming Language Standard ECMA-262.
JSONPath (JavaScript Object Notation Path)	JSONPath is XPath for JSON. It is a string syntax for selecting and extracting JSON (RFC 8259) values from within a given JSON value as defined in IETF RFC 9535.
m4	m4 [20] is a general-purpose macro processor included in most Unix-like operating systems and is a component of the POSIX standard [18].
Notifier	An element in an event notification mechanism that transmits events.
OASIS (Organization for the Advancement of Structured Information Standards)	A standards development organization that promulgates standards for data interchange.
OGC (Open Geospatial Consortium)	A standards development organization that promulgates standards for the global geospatial community.
POSIX (Portable Operating System Interface)	A set of formal descriptions that provide a standard for the design of operating systems, especially those that are compatible with UNIX. IEEE 1003, ISO/IEC 9945 [18].
Push	The process of sending an object, for example, an EIDO, from one functional element to another on an unsolicited basis.
regex	A Regular Expressions (RE) provides a mechanism to select specific strings from a set of character strings [21].

Term or Abbreviation (Expansion)	Definition / Description
SIP (Session Initiation Protocol)	A protocol specified by the IETF (RFC 3261) that defines a method for establishing multimedia sessions over the Internet. Used as the call signaling protocol in VoIP, NENA i2, and NENA i3.
Subscriber	An entity that establishes a subscription to a Notifier to be notified of information related to that subscription.
WebSocket	The WebSocket Protocol enables two-way communication between a client running untrusted code in a controlled environment to a remote host that has opted-in to communications from that code.
XPath (XML Path Language)	A query language for selecting nodes from an XML document.

7 References

- [1] International Standards Organization. *Geographic information — Geography Markup Language (GML) — Part 1: Fundamentals*. [ISO 19136-1:2020](#), January 2020.
- [2] Internet Engineering Task Force. *GEOPRIV Presence Information Data Format Location Object (PIDF-LO) Usage Clarification, Considerations, and Recommendations*. J. Winterbottom, M. Thomson, H. Tschofenig. [RFC 5491](#), March 2009.
- [3] Internet Engineering Task Force. *Specifying Civic Address Extensions in the Presence Information Data Format Location Object (PIDF-LO)*. J. Winterbottom, M. Thomson, R. Barnes, B. Rosen, and R. George. [RFC 6848](#), January 2013.
- [4] Open Geospatial Consortium Inc. *GML 3.1.1 PIDF-LO Shape Application Schema for use by the Internet Engineering Task Force (IETF)*. C. Reed and M. Thompson. [OGC 06-142r1](#), April 2007.
- [5] Internet Engineering Task Force. *Key words for use in RFCs to Indicate Requirement Levels*. S. Bradner. [RFC 2119](#), March 1997.
- [6] Internet Engineering Task Force. *Uniform Resource Identifier (URI): Generic Syntax*. T. Berners-Lee, R. Fielding and L. Masinter. [RFC 3986](#), January 2005.
- [7] Internet Engineering Task Force. *The Base16, Base32, and Base64 Data Encodings*. S. Josefsson. [RFC 4648](#), October 2006.
- [8] Internet Engineering Task Force. *An Extensible Markup Language (XML)-Based Format for Event Notification Filtering*. H. Khartabil, E. Leppanen, M. Lonnfors, and J. Costa-Requena. [RFC 4661](#), September 2006.

- [9] Internet Engineering Task Force. *Session Initiation Protocol (SIP) Event Notification Extension for Notification Rate Control*. A. Niemi, K. Kiss, and S. Loreto. [RFC 6446](#), January 2012.
- [10] Internet Engineering Task Force. *The WebSocket Protocol*. I. Fette and A. Melnikov. [RFC 6455](#), December 2011.
- [11] Internet Engineering Task Force. *Hypertext Transfer Protocol (HTTP/1.1): Semantics and Content*. R. Fielding, Ed. and J. Reschke, Ed. [RFC 7231](#), June 2014.
- [12] Internet Assigned Numbers Authority. *IANA Emergency Registry*.
- [13] National Emergency Number Association. *NENA i3 Standard for Next Generation 9-1-1*. [NENA-STA-010.3-2021](#), October 2021.
- [14] National Emergency Number Association. *NENA Standard for Emergency Incident Data Object (EIDO)*. [NENA-STA-021.1-2022](#), October 2021.
- [15] Open Geospatial Consortium. *Geography Markup Language*. [GML](#), January 2020.
- [16] Organization for the Advancement of Structured Information Standards. *Emergency Data Exchange Language Resource Messaging (EDXL-RM) 1.0*. E. Jones, P. Aymond, R. Brooks, T. Grapes, G. Ham, R. Iannella, K. Robinson, W. Joerg, and A. Triglia. [OASIS EDXL-RM](#), December 2009.
- [17] Organization for the Advancement of Structured Information Standards. *Emergency Data Exchange Language (EDXL) Distribution Element Version 2.0*. E. Jones and J. Waters. [EDXL-DE](#), September 2013.
- [18] Institute of Electrical and Electronics Engineers. *IEEE/Open Group 1003.1-2017 - IEEE Standard for Information Technology--Portable Operating System Interface (POSIX(TM)) Base Specifications, Issue 7*. A. Josey. [1003.1-2017](#), January 2018.
- [19] The Open Group. *expr - evaluate arguments as an expression*. A. Josey. [expr](#), January 2018.
- [20] The Open Group. *m4 - macro processor*. A. Josey. [m4](#), January 2018.
- [21] The Open Group. *Regular Expressions*. A. Josey. [REs](#), January 2018.
- [22] Internet Engineering Task Force. *JSONPath: Query Expressions for JSON*. S. Gössner, G. Normington, C. Bormann. [RFC 9535](#), February 2024.
- [23] National Emergency Number Association. *NENA Standard for WebSocket-based Subscriptions and Notifications between Next Generation (NG9-1-1) Systems and Applications*. [Version 1.1 Schema](#). 2023.
- [24] National Emergency Number Association. *NENA Standard for NG9-1-1 Logging Service*. [Version 1.2 Schema](#). 2022.
- [25] National Emergency Number Association. *NENA Standard for NG9-1-1 EIDO Conveyance Incident Data API*. [Version 1.0 Schema](#). 2022.
- [26] OpenAPI Initiative. *OpenAPI Specification 3.1.0 Released*. <https://www.openapis.org/blog/2021/02/18/openapi-specification-3-1-released>, February 18, 2021.

- [27] Internet Engineering Task Force. *Filtering Location Notifications in the Session Initiation Protocol (SIP)*. R. Mahy, B. Rosen, and H. Tschofenig. [RFC 6447](#), January 2012.

8 Exhibit

Not Applicable.

9 Appendix A: Subscription Filter Examples

9.1 QualFilter Example

Multnomah County, Oregon is across the Columbia River from Clark County, Washington. In addition to the part of the Columbia River they share, there are two Interstate bridges across the river between the two counties, the I5 bridge and the I205 bridge.

In this example, the Clark County PSAP wants to receive some EIDOs from Multnomah County. Since they don't want to receive all of Multnomah County's EIDOs, they will configure a filter that Multnomah County will use to test each EIDO to see whether or not to send it to Clark County.

The filter will specify that the only EIDOs that Clark County wants to receive are those that are related to Columbia River emergency incidents or incidents on either of the Interstate bridges that may affect Clark County.

To implement this filter, an expression is needed to compare the EIDO element values with the filter to determine if there is a match. Any EIDO that matches the filter will be sent to the subscriber.

If this logical expression is "true" when the filter is compared to an EIDO, it is considered a match and the EIDO would be sent:

(EIDO River Location) and ((River Incident Type) or (River Note Keyword))

or

(EIDO I5 Bridge Location) and ((Bridge Incident Type) or (Bridge Note Keyword))

or

(EIDO I205 Bridge Location) and ((Bridge Incident Type) or (Bridge Note Keyword))

Where:

EIDO River Location is true when:

Any of these elements in the EIDO are within the PIDF of the area encompassing the Columbia River for the area of the river shared by both counties:

Incident Location	EIDO:Incident/EIDO:Location
Caller Location	EIDO:Call/EIDO:Location
Unit Location	EIDO:EmergencyResource/EIDO:UnitLocation

River Incident Type is true when the Incident Type Common EIDO element (EIDO:Incident/EIDO:IncidentTypeCommonRegistryText) contains at least one of the following values:

WATER (Water Rescue)
RESCUE (Rescue)
MISBOATR (Missing Boater)
MCI (Mass Casualty Incident)
MAYDAY (Mayday)
FMARINE (Fire Marine)
EXPLOSION (Explosion)
DROWN (Drowning)
DEATH (Death)
BRGCLPSE (Bridge Collapse)

River Note Keyword is true if any of these EIDO notes fields:

EIDO Notes (Header)	EIDO:Notes/EIDO:NotesActionComments
Additional Data Notes	EIDO:AdditionalData/EIDO:Notes
Incident Location Notes	EIDO:Location/EIDO:Notes

Include any of these keywords:

River
Drown
Drowning
Boat

EIDO I5 Bridge Location is true when:

Any of these elements in the EIDO are within the PIDF of the area encompassing the I5 bridge:

Incident Location	EIDO:Incident/EIDO:Location
Caller Location	EIDO:Call/EIDO:Location
Unit Location	EIDO:EmergencyResource/EIDO:UnitLocation

Bridge Incident Type is true when the Incident Type Common EIDO element (EIDO:Incident/EIDO:IncidentTypeCommonRegistryText) contains at least one of the following values:

TSTOP (Traffic Stop)

VEHABND (Abandoned Vehicle)
THAZ (Traffic Hazard)
PURSUIT (Pursuit)
MCI (Mass Casualty Incident)
FVEH (Fire Vehicle)
EXPLOSION (Explosion)
DISVEH (Disabled Vehicle)
DEATH (Death)

Bridge Note Keyword is true if any of these EIDO notes fields:

EIDO Notes (Header)	EIDO:Notes/EIDO:NotesActionComments
Additional Data Notes	EIDO:AdditionalData/EIDO:Notes
Incident Location Notes	EIDO:Location/EIDO:Notes

Include any of these keywords:

Crash
Wreck
Accident
Fire

EIDO I205 Bridge Location is true when:

Any of these location elements in the EIDO are within the polygon of the area encompassing the I205 Bridge:

Incident Location	EIDO:Incident/EIDO:Location
Caller Location	EIDO:Call/EIDO:Location
Unit Location	EIDO:EmergencyResource/EIDO:UnitLocation

I205 Bridge GML

```
<?xml version="1.0" encoding="utf-8"?>
<Polygon xmlns="http://www.opengis.net/gml"
srsName="urn:ogc:def:crs:EPSG::4326">
  <exterior>
    <LinearRing>
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  </exterior>
</Polygon>
```

</Polygon>

I5 Bridge I5 GML

```
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<Polygon xmlns="http://www.opengis.net/gml"
srsName="urn:ogc:def:crs:EPSG::4326">
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      <posList>
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  </exterior>
</Polygon>
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Columbia River GML

```
<?xml version="1.0" encoding="utf-8"?>
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srsName="urn:ogc:def:crs:EPSG::4326">
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```

ACKNOWLEDGMENTS

The National Emergency Number Association (NENA) 9-1-1 Core Services Committee, EIDO Conveyance Working Group developed this document.

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